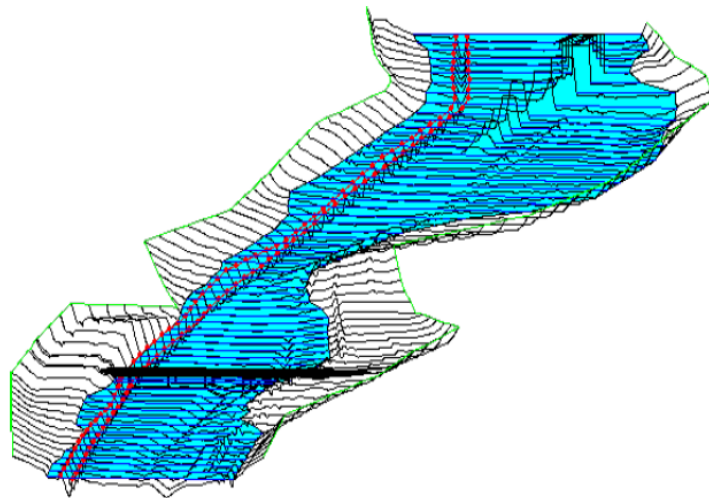


Topic	Surface Water Modelling	
Module	2.	Hydraulic Modelling (steady/unsteady)



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Learning objectives: After completion of this module the learner will be able to

1. explain the principles and objectives of hydraulic modelling (Level 2: Understand)
2. explain the structure and variables of a hydraulic model (Level 2: Understand)
3. explain the main types of hydraulic models (Level 2: Understand)
4. explain the steps to develop a hydraulic model (Level 2: Understand)
5. develop a simple hydraulic model (Level 6: Create)
6. select model parameters (Level 5: Evaluate)
7. develop a hydraulic model (Level 3: Apply) to compare and contrast the effect of implementation of various infrastructure measures on flow properties (Level 4: Analyse)

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STEADY VS UNSTEADY FLOW WITH HEC-RAS

BASIC DATA REQUIREMENTS

1 OVERVIEW

Objectives

Hydraulic modelling systems are based on the fundamental equations of fluid mechanics: conservation of mass, energy and momentum. Here the equations used in the modelling system HEC-RAS (Hydrologic Engineering Center – River Analysis System) for 1-dimensional (1D) hydrodynamic modelling will be reviewed, and their application will be discussed.

Hydraulic modelling with software systems such as HEC-RAS spans a wide range of engineering applications. Examples are the determination of

- flow depth in rivers, canals and other conveyance conduits,
- changes in flow depth due to channel controls, such as weirs, gates and spillways,
- changes in river stage during flood events,
- optimal channel design,
- stream restoration
- outflow from reservoirs,
- flow across roads during floods,
- water infrastructure design parameters,
- sediment transport, and
- many others.

General Philosophy of the HEC-RAS Modelling System

HEC-RAS is an integrated system of software, designed for interactive use in a multi-tasking, multiuser network environment. The system is comprised of a graphical user interface (GUI), separate hydraulic analysis components, data storage and management capabilities, graphics and reporting facilities. The HEC-RAS system contains four one-dimensional river analysis components for: (1) steady flow water surface profile computations; (2) unsteady flow simulation (one-dimensional and two-dimensional hydrodynamics); (3) Quasi unsteady or fully unsteady flow movable boundary sediment transport computations (1D and 2D); and (4) water quality analysis. A key element is that all four components use a common geometric data representation and common geometric and hydraulic computation routines. In addition to the four river analysis components, the system contains several hydraulic design features that can be invoked once the basic water surface profiles are computed. HEC-RAS also has an extensive spatial data integration and mapping system (HEC-RAS Mapper). The current version of HEC-RAS supports Steady and Unsteady flow water surface profile calculations; combined 1D and 2D hydrodynamics; sediment transport/mobile bed computations; water temperature analysis; water quality analyses (nutrient transport and fate); and spatial mapping of many computed parameters (depth, water surface elevation, velocity, etc.). Note that HEC-RAS is continuously improved by the US Army Corps of Engineers, and new features and additional capabilities will be added in future releases.

Overview of Hydraulic Capabilities of HEC-RAS

HEC-RAS is designed to perform one-dimensional (1D), two-dimensional (2D), or combined 1D and 2D hydraulic calculations for a full network of natural and constructed channels. The following is a description of the major hydraulic capabilities of HEC-RAS.

Steady Flow Water Surface Profiles. This component of the modelling system is intended for calculating water surface profiles for steady gradually varied flow. The system can handle a single river reach, a dendritic system, or a full network of channels. The steady flow

component is capable of modelling subcritical, supercritical, and mixed flow regime water surface profiles. The basic computational procedure is based on the solution of the one-dimensional energy equation. Energy losses are evaluated by friction (Manning's equation) and contraction/expansion (coefficient multiplied by the change in velocity head). The momentum equation is utilized in situations where the water surface profile is rapidly varied. These situations include mixed flow regime calculations (i.e., hydraulic jumps), hydraulics of bridges, and evaluating profiles at river confluences (stream junctions). The effects of various obstructions such as bridges, culverts, weirs, spillways and other structures in the flood plain may be considered in the computations. The steady flow system is designed for application in flood plain management and flood insurance studies to evaluate floodway encroachments. Also, capabilities are available for assessing the change in water surface profiles due to channel improvements, and levees (i.e. stopbanks). Special features of the steady flow component include: multiple plan analyses; multiple profile computations; multiple bridge and/or culvert opening analysis, and split flow optimisation at stream junctions and lateral weirs and spillways.

Unsteady Flow Simulation. This component of the HEC-RAS modelling system is capable of simulating one-dimensional unsteady flow; two-dimensional unsteady flow; or combined 1D and 2D unsteady flow modelling through a full network of open channels. The 1D unsteady flow equation solver was adapted from Dr. Robert L. Barkau's UNET model (Barkau, 1992 and HEC, 1997). This 1D unsteady flow component was developed primarily for subcritical flow regime calculations. The 2D unsteady flow equation solver was developed at HEC and was directly integrated into the HEC-RAS Unsteady flow engine in order to facilitate combined 1D and 2D hydrodynamic modelling. The hydraulic calculations for cross-sections, bridges, culverts, and other hydraulic structures that were developed for the steady flow component were incorporated into the unsteady flow module. Additionally, the unsteady flow component has the ability to model storage areas and hydraulic connections between storage areas; 2D Flow Areas; and between stream reaches.

Sediment Transport/Movable Boundary Computations. This component of the modelling system is intended for the simulation of one-dimensional and two-dimensional sediment transport/movable boundary calculations resulting from scour and deposition over moderate time periods (typically years, although applications to single flood events will be possible). The sediment transport potential is computed by grain size fraction, thereby allowing the simulation of hydraulic sorting and armouring. Major features include the ability to model a full network of streams, channel dredging, various levee and encroachment alternatives, and the use of several different equations for the computation of sediment transport. The model is designed to simulate long-term trends of scour and deposition in a stream channel that might result from modifying the frequency and duration of the water discharge and stage, or modifying the channel geometry. This system can be used to evaluate deposition in reservoirs, design channel contractions required to maintain navigation depths, predict the influence of dredging on the rate of deposition, estimate maximum possible scour during large flood events, and evaluate sedimentation in fixed channels.

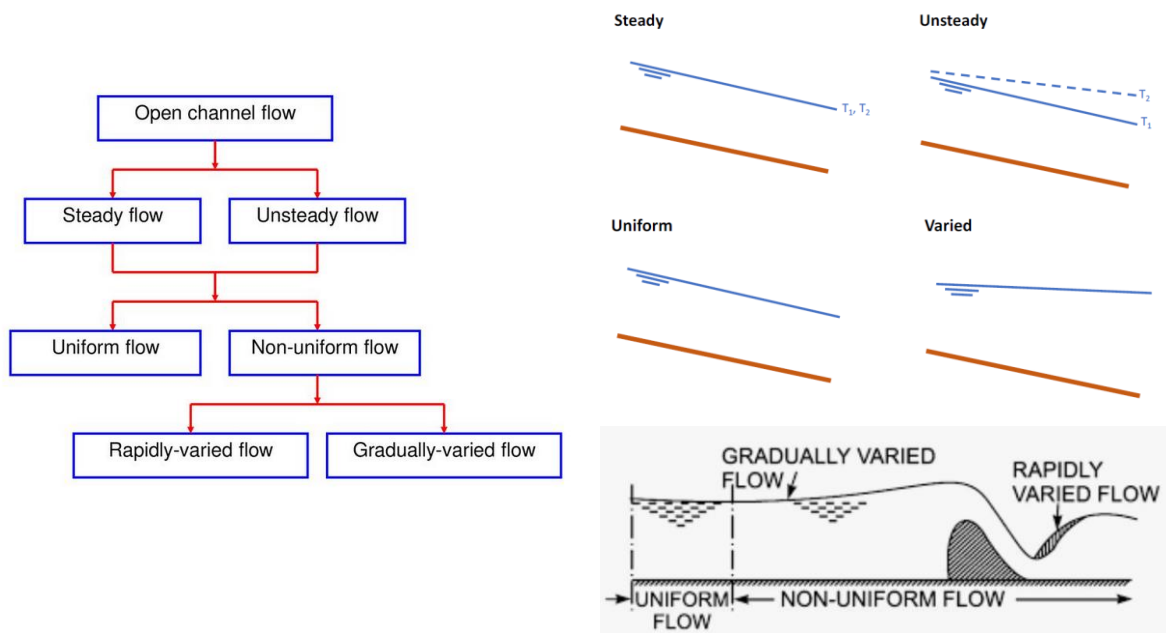
Water Quality Analysis. This component of the modelling system is intended to allow the user to perform riverine water quality analyses. The current version of HEC-RAS can perform detailed temperature analysis and transport of a limited number of water quality constituents (Algae, Dissolved Oxygen, Carbonaceous Biological Oxygen Demand, Dissolved Orthophosphate, Dissolved Organic Phosphorus, Dissolved Ammonium Nitrate, Dissolved Nitrite Nitrogen, Dissolved Nitrate Nitrogen, and Dissolved Organic Nitrogen). Future versions

of the software will include the ability to perform the transport of several additional water quality constituents.

2 THEORETICAL BASIS FOR STEADY ONE-DIMENSIONAL HYDRODYNAMIC CALCULATIONS

This chapter describes the methodologies used in performing the one-dimensional (1D) steady flow and unsteady flow calculations. Two-dimensional (2D) unsteady flow calculations are covered elsewhere. The basic equations are presented along with discussions of the various terms. Solution schemes for the various equations are described. Discussions are provided as to how the equations should be applied, as well as applicable limitations.

Brief review: Classification of open channel flow



1D Steady Flow Water Surface Profiles

HEC-RAS is currently capable of performing 1D water surface profile calculations for steady gradually varied flow in natural or constructed channels. Subcritical, supercritical, and mixed flow regime water surface profiles can be calculated. Topics discussed in this section include: equations for basic profile calculations; cross section subdivision for conveyance calculations; composite Manning's n for the main channel; evaluation of the mean kinetic energy head (velocity weighting coefficient α); friction loss evaluation; contraction and expansion losses; computational procedure; critical depth determination; applications of the momentum equation; air entrainment in high velocity streams; and limitations of the steady flow model.

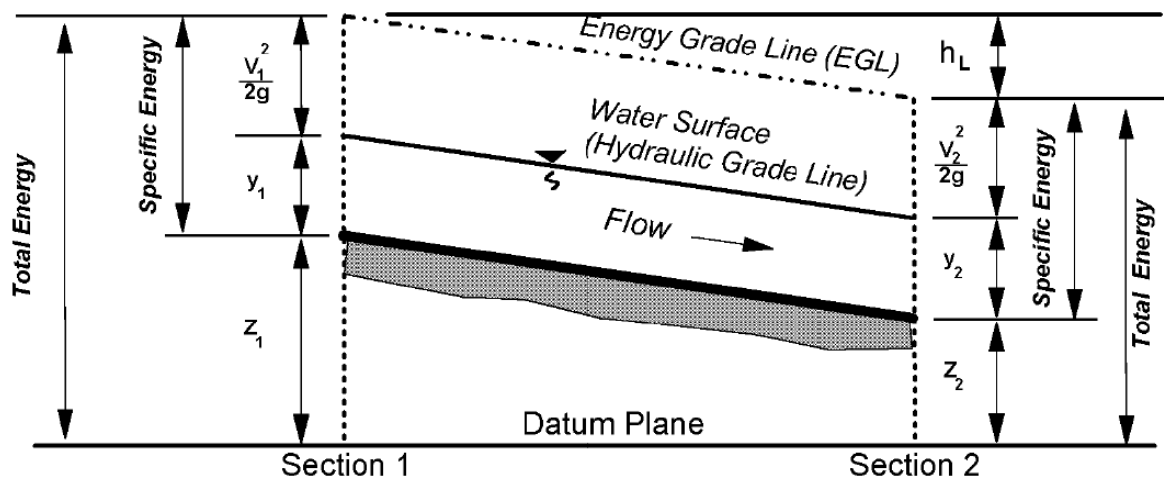
Equations for Basic Profile Calculations

Water surface profiles are computed from one cross section to the next by solving the Energy equation with an iterative procedure called the *standard step method*. The Energy equation is

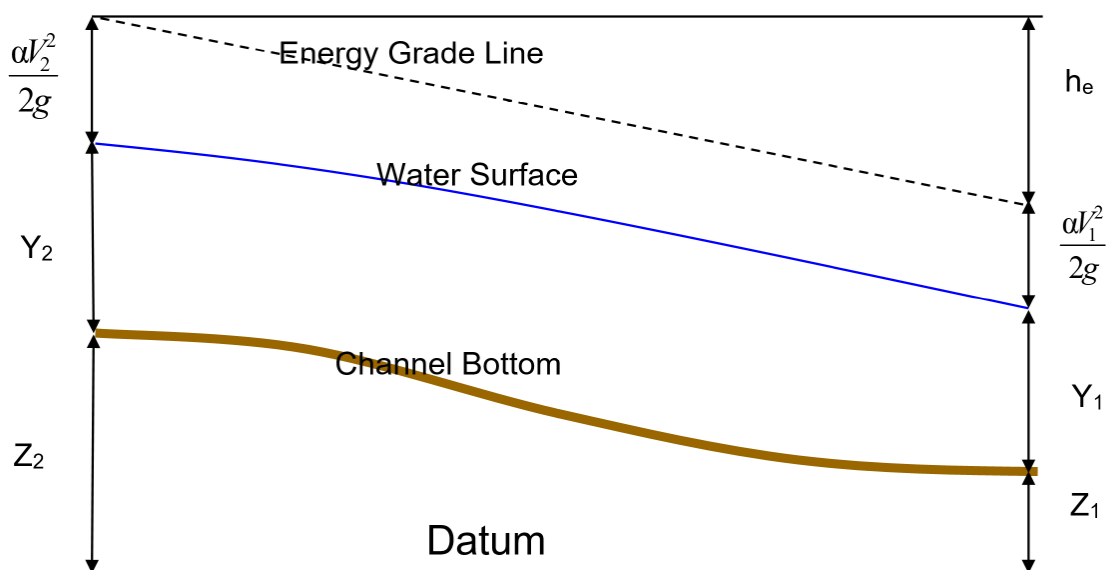
$$Z_2 + Y_2 + \frac{a_2 V_2^2}{2g} = Z_1 + Y_1 + \frac{a_1 V_1^2}{2g} + h_e \quad (1)$$

Symbol	Description
Z_1, Z_2	elevation of the main channel inverts
Y_1, Y_2	depth of water at cross sections
V_1, V_2	average velocities (total discharge/ total flow area)
a_1, a_2	velocity weighting coefficients
g	gravitational acceleration
h_e	energy head loss

A diagram showing the terms of the energy equation is shown in the figure below (in this figure the head loss is termed h_L), with differentiation between total energy and specific energy.



Considering the kinetic energy correction factor α :



The energy head loss (h_e) between two cross sections is comprised of friction losses and contraction or expansion losses. The equation for the energy head loss is as follows:

$$h_e = L\bar{S}_f + C \left| \frac{a_2 V_2^2}{2g} - \frac{a_1 V_1^2}{2g} \right| \quad (2)$$

Symbol	Description
L	discharge weighted reach length
$\bar{S}_f$	representative friction slope between two sections
C	expansion or contraction loss coefficient

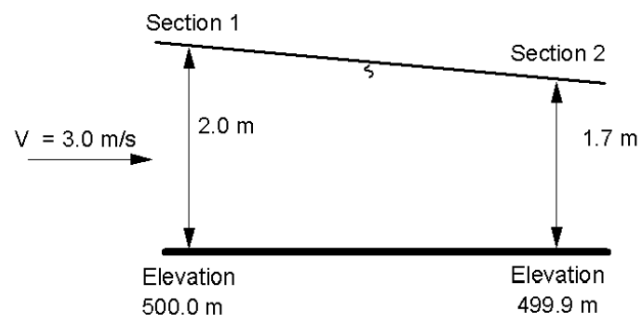
The distance weighted reach length, L , is calculated as:

$$L = \frac{L_{lob} \bar{Q}_{lob} + L_{ch} \bar{Q}_{ch} + L_{rob} \bar{Q}_{rob}}{\bar{Q}_{lob} + \bar{Q}_{ch} + \bar{Q}_{rob}} \quad (3)$$

Symbol	Description
L_{lob}, L_{ch}, L_{rob}	cross section reach lengths specified for flow in the left overbank, main channel, and right overbank, respectively
$\bar{Q}_{lob} + \bar{Q}_{ch} + \bar{Q}_{rob}$	arithmetic average of the flows between sections for the left overbank, main channel, and right overbank, respectively

Exercise 1

The velocity of the upstream end of a rectangular channel 1 m wide is 3.0 m/s, and the flow depth is 2.0 m. The depth at the downstream end is 1.7 m. The elevation at section 1 is 500 m and at 2 is 499.90 m. Determine the headloss due to friction. Assume the kinetic energy correction factor is 1.0.



Cross Section Subdivision for Conveyance Calculations

The determination of total conveyance and the velocity coefficient for a cross section requires that flow be subdivided into units for which the velocity is uniformly distributed. The approach used in HEC-RAS is to subdivide flow in the **overbank** areas using the input cross section n-value break points (locations where n-values change) as the basis for subdivision

(see Figure below). Conveyance is calculated within each subdivision from the following form of Manning's equation:

$$Q = KS_f^{1/2} \quad (4)$$

$$K = \frac{a}{n} AR^{2/3} \quad (5)$$

Where in Equation (5) $a = 1$ for SI units, and $a = 1.486$ for English units.

Symbol	Description	Units
K	conveyance for subdivision	
n	Manning's roughness coefficient for subdivision	
A	flow area for subdivision	
R	hydraulic radius for subdivision (area / wetted perimeter)	
S_f	slope of the energy gradeline	

It is convenient to discuss channel capacity in this manner, since channel slopes are usually fairly fixed by the natural topography of the land surface. On the other hand, engineers have some control over the factors that make up the conveyance. Channel capacities for various discharges can be computed by plotting K versus depth for given cross sections. The conveyance can also be used in cases of compound channel sections to balance the discharge between the different channel components, since the slopes of all components will be the same.

Flood control channels and water delivery canals are usually designed to carry a specified discharge at normal depth. Thus, the relationship among discharge, depth, and channel geometry embodied in Equation (4) is central to the practice of open channel hydraulics. Design situations are normally presented in one of two cases: (a) The channel geometry and slope are known and the normal depth is to be calculated, or (b) channel dimensions are to be determined to carry a given discharge at a specified normal depth.

Brief review: Types of open channels

Prismatic and non-prismatic channels

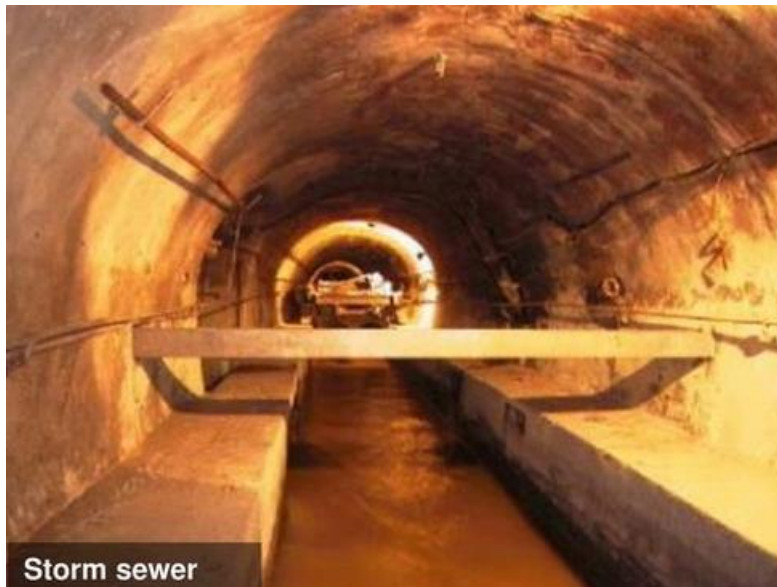
Prismatic channels are channels with constant cross-sectional shape, size and bottom slope. Most man-made (artificial) channels are prismatic over long stretches. Examples of man-made channels are irrigation canals, flumes, drainage ditches, roadside channels, drops, chutes, culverts, among others. Examples are shown below.



Natural channels have in general varying cross-sections and are therefore non-prismatic. Examples are hillside rivulets, through brooks, streams, rivers or tidal estuaries.

Rigid and mobile boundary channels

Rigid channels are channels with non-deformable boundaries. Channel geometry and roughness are constant over time. Typical examples are lined canals, sewers and non-erodible unlined canals.



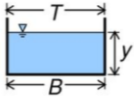
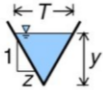
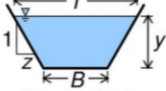
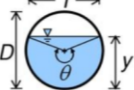
Storm sewer

Mobile boundary channels are channels with boundaries that undergo deformation due to the continuous process of erosion and deposition due to the flow. Examples are man-made channels built from erodible material or natural rivers. Examples are shown below.





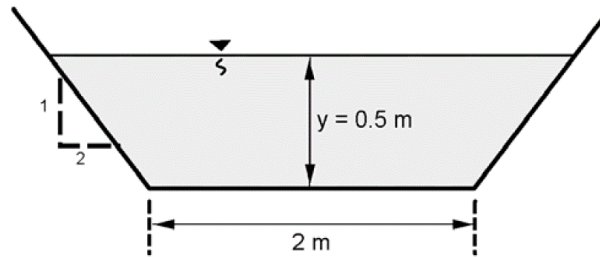
In the table below selected open channel geometry parameters are summarised for a range of channel shapes.

Channel shape	Area A	Top width T	Wetted perimeter P
 Rectangular	By	B	$B + 2y$
 Triangular	zy^2	$2zy$	$2y\sqrt{1+z^2}$
 Trapezoidal	$By + zy^2$	$B + 2zy$	$B + 2y\sqrt{1+z^2}$
 Circle	$\frac{D^2}{8}(\theta - \sin\theta)$ θ in radian	$D\left(\frac{\sin\theta}{2}\right)$ θ in angle	$\frac{\theta D}{2}$ θ in radian

Exercise 2

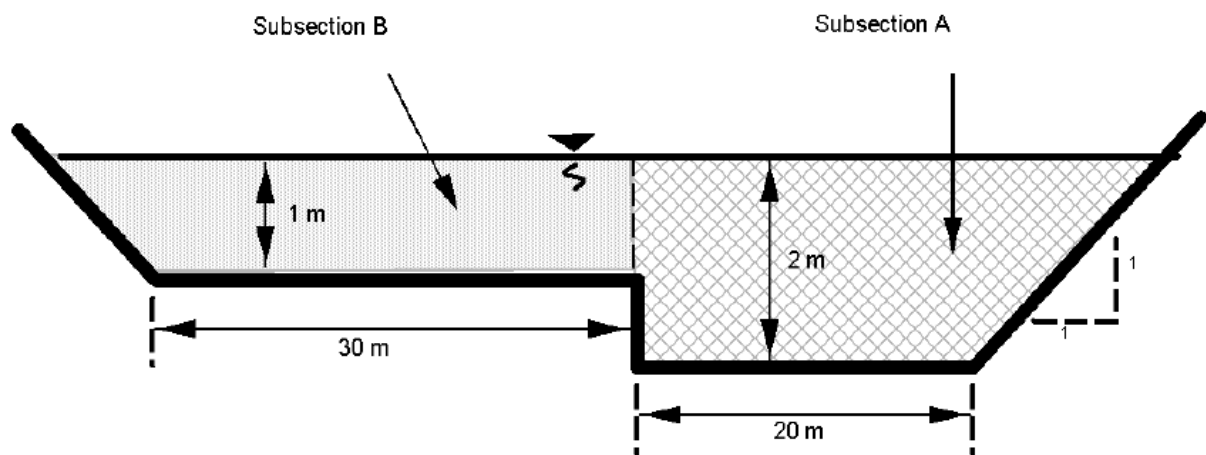
(a) A rectangular channel 5 m wide carries a discharge of 100 m³/s at normal (uniform flow) depth. The channel is running on a slope of 0.005 m/m and has a Manning n value of 0.035 (metric). Determine the normal depth (uniform flow depth) for this situation.

(b) Determine the velocity (V) and discharge (Q) for the trapezoidal earth channel shown below, with $B = 2$ m, sideslope $1V:2H$, $S = 0.003$ m/m, normal (uniform) depth $y = 0.5$ m, and $n = 0.02$ (metric).



Exercise 3

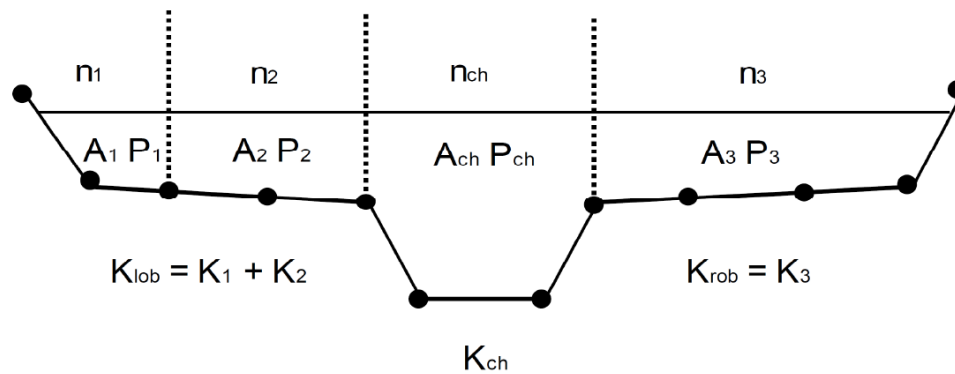
A compound channel as illustrated, with an n value of 0.03 (metric), a longitudinal slope of 0.002 m/m and sideslopes of $1V:1H$.



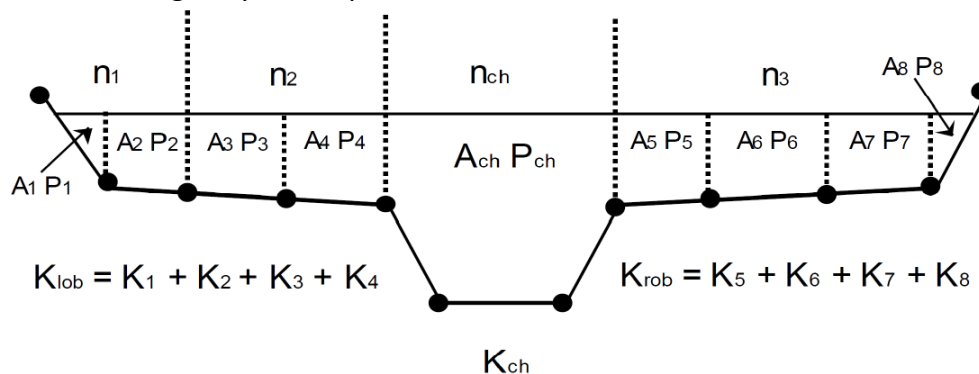
Determine the discharge Q (i) with and (ii) without subdividing the channel and compare the results.

Calculate and plot the conveyance curve for this cross section and calculate the total discharge. Note: With a subdivided cross section, the conveyance at a given flow depth should be calculated for each subsection, and then added together to get the total conveyance at that depth.

In HEC-RAS all the incremental conveyances in the overbanks are summed up to obtain a conveyance for the left overbank and the right overbank. The main channel conveyance is normally computed as a single conveyance element. The total conveyance for the cross section is obtained by summing the three subdivision conveyances (left, channel, and right).



An alternative method available in HEC-RAS is to calculate conveyance between every coordinate point in the overbanks (see Figure below). The conveyance is then summed to get the total left overbank and right overbank values. This method is used in the Corps HEC-2 program. The method has been retained as an option within HEC-RAS in order to reproduce studies that were originally developed with HEC-2.

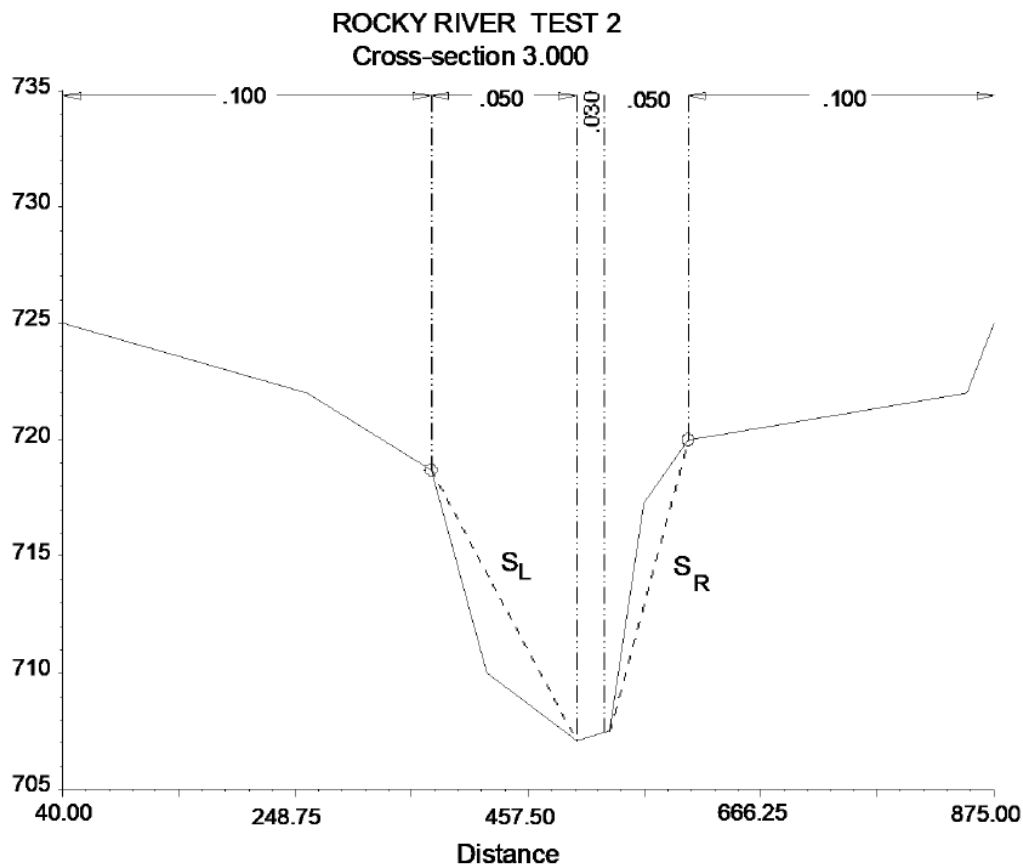


The two methods for computing conveyance will produce different answers whenever portions on the overbank have ground sections with significant vertical slopes. In general, the HEC-RAS default approach will provide a lower total conveyance for the same water surface elevation. In order to test the significance of the two ways of computing conveyance, comparisons were performed using 97 data sets from the HEC profile accuracy study (HEC, 1986). Water surface profiles were computed for the 1% chance event using the two methods for computing conveyance in HEC-RAS. The results of the study showed that the HEC-RAS default approach will generally produce a higher computed water surface elevation. Out of the 2048 cross section locations, 47.5% had computed water surface elevations within 30.48 mm, 71% within 60.96 mm, 94.4% within 121.92 mm, 99.4% within 304.8 mm, and one cross section had a difference of 0.84 m. Because the differences tend to be in the same direction, some effects can be attributed to propagation of downstream differences. The results from the conveyance comparisons do not show which method is more accurate, they only show differences. In general, it is felt that the HEC-RAS default method is more commensurate with the Manning equation and the concept of separate flow elements. Further research, with observed water surface profiles, will be needed to make any conclusions about the accuracy of the two methods.

Composite Manning's n for the Main Channel

Flow in the main channel is not subdivided, except when the roughness coefficient is changed within the channel area. HEC-RAS tests the applicability of subdivision of roughness within the main channel portion of a cross section, and if it is not applicable, the program will

compute a single composite n value for the entire main channel. The program determines if the main channel portion of the cross section can be subdivided or if a composite main channel n value will be utilized based on the following criterion: if a main channel side slope is steeper than 5H:1V and the main channel has more than one n -value, a composite roughness n_c will be computed [Equation 6-17 in Chow, 1959]. The channel side slope used by HEC-RAS is defined as the horizontal distance between adjacent n -value stations within the main channel over the difference in elevation of these two stations (See S_L and S_R of the figure below).



For the determination of n_c , the main channel is divided into N parts, each with a known wetted Perimeter P_i and roughness coefficient n_i .

$$n_c = \left[\frac{\sum_{i=1}^N (P_i n_i^{1.5})}{P} \right]^{2/3} \quad (6)$$

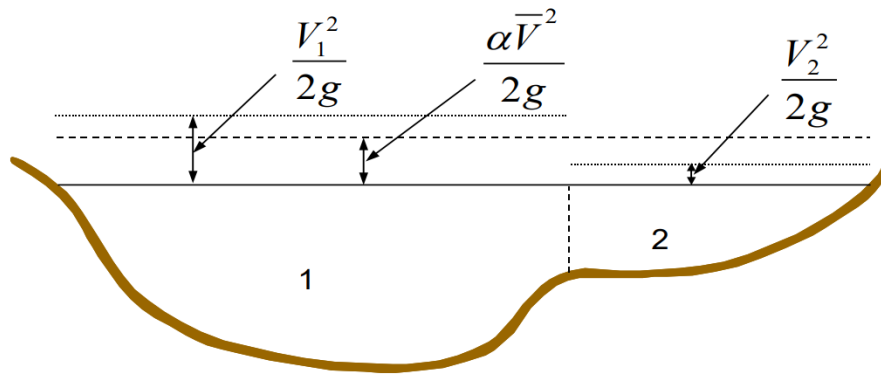
Symbol	Description
n_c	composite or equivalent coefficient of roughness
P	wetted perimeter of entire main channel

Symbol	Description
P_i	wetted perimeter of subdivision i
n_i	coefficient of roughness for subdivision

The computed composite n_c should be checked for reasonableness. The computed value is the composite main channel n value in the output and summary tables.

Evaluation of the Mean Kinetic Energy Head

Within the 1D river reach segments, only a single water surface and therefore a single mean energy are computed at each cross section. For a given water surface elevation, the mean energy is obtained by computing a flow weighted energy from the three subsections of a cross section (left overbank, main channel, and right overbank). The figure below shows how the mean energy would be obtained for a cross section with a main channel and a right overbank (no left overbank area).



V_1 = mean velocity for subarea 1

V_2 = mean velocity for subarea 2

To compute the mean kinetic energy it is necessary to obtain the velocity head weighting coefficient α (also called kinetic energy correction factor or velocity coefficient). The velocity head weighting coefficient is used to account for the nonuniformity of the velocity distribution when using the average velocity to compute the kinetic energy term. Thus, it is a function of the channel geometry, roughness and slope. It is important to assess the given hydrodynamic situation to identify the correct value for α . Albeit a value of 1 is used under the assumption of uniformity in the velocity distribution, the factor α can vary from about 1.05 for fairly uniform cross sections up to a value of 1.2 for very nonuniform sections. A common way to calculate α (also implemented in HEC-RAS) is as follows:

Mean Kinetic Energy Head = Discharge-Weighted Velocity Head

$$\alpha \frac{\bar{V}^2}{2g} = \frac{Q_1 \frac{V_1^2}{2g} + Q_2 \frac{V_2^2}{2g}}{Q_1 + Q_2} \quad (7)$$

$$\alpha = \frac{2g \left(Q_1 \frac{V_1^2}{2g} + Q_2 \frac{V_2^2}{2g} \right)}{(Q_1 + Q_2) \bar{V}^2} \quad (8)$$

$$\alpha = \frac{Q_1 V_1^2 + Q_2 V_2^2}{(Q_1 + Q_2) \bar{V}^2} \quad (9)$$

$$\alpha = \frac{[Q_1 V_1^2 + Q_2 V_2^2 + \dots Q_N V_N^2]}{Q \bar{V}^2} \quad (10)$$

The velocity coefficient, α , is computed based on the conveyance in the three flow elements: left overbank, right overbank, and channel. It can also be written in terms of conveyance and area as in the following equation

$$\alpha = \frac{(A_t)^2 \left[\frac{K_{lob}^3}{A_{lob}^2} + \frac{K_{ch}^3}{A_{ch}^2} + \frac{K_{rob}^3}{A_{rob}^2} \right]}{K_t^3} \quad (11)$$

Symbol	Description
A_t	total flow area of cross section
A_{lob}, A_{ch}, A_{rob}	flow areas of left overbank, main channel and right overbank, respectively
K_t	total conveyance of cross section
K_{lob}, K_{ch}, K_{rob}	conveyances of left overbank, main channel and right overbank

Friction Loss Evaluation

Friction loss is evaluated in HEC-RAS as the product of $\bar{S}_f$ and L (Equation 2), where $\bar{S}_f$ is the representative friction slope for a reach, and L is defined by Equation (3). The friction slope (slope of the energy grade line) at each cross section is computed from Manning's equation as follows:

$$S_f = \left(\frac{Q}{K} \right)^2 \quad (12)$$

Alternative expressions for the representative reach friction slope in HEC.RAS are as follows:
Average Conveyance Equation

$$\overline{S}_f = \left(\frac{Q_1 + Q_2}{K_1 + K_2} \right)^2 \quad (13)$$

Average Friction Slope Equation

$$\overline{S}_f = \frac{S_{f1} + S_{f2}}{2} \quad (14)$$

Geometric Mean Friction Slope Equation

$$\overline{S}_f = \sqrt{S_{f1} \times S_{f2}} \quad (15)$$

Harmonic Mean Friction Slope Equation

$$\overline{S}_f = \frac{2 (S_{f1} \times S_{f2})}{S_{f1} + S_{f2}} \quad (16)$$

Where K_1 and K_2 are the conveyances at the upstream and downstream sections, and S_{f1} and S_{f2} are the friction slopes at the upstream and downstream sections, respectively. The Average Conveyance method (Equation 13) is the "default" equation used by the program; that is, it is used automatically unless a different equation is selected by the user. The program also contains an option to select equations, depending on flow regime and profile type (e.g., S1, M1, etc.). Further discussion of the alternative methods for evaluating friction loss is contained in the HEC-RAS Hydraulic Reference Manual.

Contraction and Expansion Loss Evaluation

Contraction and expansion losses in HEC-RAS are evaluated by the following equation:

$$h_{ce} = C \left| \frac{\alpha_1 V_1^2}{2g} - \frac{\alpha_2 V_2^2}{2g} \right| \quad (17)$$

Where: C = the contraction or expansion coefficient

The program assumes that a contraction is occurring whenever the velocity head downstream is greater than the velocity head upstream. Likewise, when the velocity head upstream is greater than the velocity head downstream, the program assumes that a flow expansion is occurring. Typical C values can be found in the section "Basic Data Requirements".

Computation Procedure

The unknown water surface elevation at a cross section is determined by an iterative solution of Equation (1) and Equation (2). The computational procedure is as follows:

1. Assume a water surface elevation at the upstream cross section (or downstream cross section if a supercritical profile is being calculated).

2. Based on the assumed water surface elevation, determine the corresponding total conveyance and velocity head.
3. With values from step 2, compute $\bar{S}_f$ and solve Equation (2) for h_e .
4. With values from steps 2 and 3, solve Equation (1) for WS2.
5. Compare the computed value of WS2 with the value assumed in step 1; repeat steps 1 through 5 until the values agree to within 0.003 m (0.01 feet), or the user-defined tolerance.

The criterion used to assume water surface elevations in the iterative procedure varies from trial to trial. The first trial water surface is based on projecting the previous cross section's water depth onto the current cross section. The second trial water surface elevation is set to the assumed water surface elevation plus 70% of the error from the first trial (computed W.S. - assumed W.S.). In other words, W.S. new = W.S. assumed + 0.70 * (W.S. computed - W.S. assumed). The third and subsequent trials are generally based on a "Secant" method of projecting the rate of change of the difference between computed and assumed elevations for the previous two trials. The equation for the secant method is as follows:

$$WS_I = WS_{I-1} - Err_{I-1} \times \frac{Assum_Diff}{Err_Diff} \quad (18)$$

Symbol	Description
WS_I	the new assumed water surface
WS_{I-1}	the previous iteration's assumed water surface
WS_{I-2}	the assumed water surface from two trials previous
Err_{I-2}	the error from two trials previous (computed water surface minus assumed from the I-2 iteration)
$Assum_Diff$	the difference in assumed water surfaces from the previous two trials. $Assum_Diff = WS_{I-2} - WS_{I-1}$
Err_Diff	the difference in the previous error (Err_{I-2}) and the current error (Err_{I-1}). $Err_Diff = Err_{I-2} - Err_{I-1}$

The change from one trial to the next is constrained to a maximum of 50 percent of the assumed depth from the previous trial. On occasion the secant method can fail if the value of Err_Diff becomes too small. If the Err_Diff is less than 1.0E-2, then the secant method is not used. When this occurs, the program computes a new guess by taking the average of the assumed and computed water surfaces from the previous iteration. The program is constrained by a maximum number of iterations (the default is 20) for balancing the water surface. While the program is iterating, it keeps track of the water surface that produces the minimum amount of error between the assumed and computed values. This water surface is called the minimum error water surface. If the maximum number of iterations is reached before a balanced water surface is achieved, the program will then calculate critical depth (if this has not already been done). The program then checks to see if the error associated with the minimum error water surface is within a predefined tolerance (the default is 0.1 m or 0.3 ft). If the minimum error water surface has an associated error less than the predefined

tolerance, and this water surface is on the correct side of critical depth, then the program will use this water surface as the final answer and set a warning message that it has done so. If the minimum error water surface has an associated error that is greater than the predefined tolerance, or it is on the wrong side of critical depth, the program will use critical depth as the final answer for the cross section and set a warning message that it has done so. The rationale for using the minimum error water surface is that it is probably a better answer than critical depth, as long as the above criteria are met. Both the minimum error water surface and critical depth are only used in this situation to allow the program to continue the solution of the water surface profile. Neither of these two answers are considered to be valid solutions, and therefore warning messages are issued when either is used. In general, when the program cannot balance the energy equation at a cross section, it is usually caused by an inadequate number of cross sections (cross sections spaced too far apart) or bad cross section data. Occasionally, this can occur because the program is attempting to calculate a subcritical water surface when the flow regime is actually supercritical. When a balanced water surface elevation has been obtained for a cross section, checks are made to ascertain that the elevation is on the right side of the critical water surface elevation (e.g., above the critical elevation if a subcritical profile has been requested by the user). If the balanced elevation is on the wrong side of the critical water surface elevation, critical depth is assumed for the cross section and a warning message to that effect is displayed by the program. The program user should be aware of critical depth assumptions and determine the reasons for their occurrence, because in many cases they result from reach lengths being too long or from misrepresentation of the effective flow areas of cross sections. For a subcritical profile, a preliminary check for proper flow regime involves checking the Froude number. The program calculates the Froude number of the balanced water surface for both the main channel only and the entire cross section. If either of these two Froude numbers are greater than 0.94, then the program will check the flow regime by calculating a more accurate estimate of critical depth using the minimum specific energy method (this method is described in the next section). A Froude number of 0.94 is used instead of 1.0, because the calculation of Froude number in irregular channels is not accurate. Therefore, using a value of 0.94 is conservative, in that the program will calculate critical depth more often than it may need to. For a supercritical profile, critical depth is automatically calculated for every cross section, which enables a direct comparison between balanced and critical elevations.

Exercise 4

Calculation of the water surface profile upstream of a dam

One purpose for which dams are designed and built is flood protection. Among the range of criteria that need to be considered, the design engineer must ascertain that the dam height is sufficient for the annual exceedance probability of the design flood, and the design engineer must also ascertain that the change in water level upstream of the dam does not result in negative consequences. You will investigate these two aspects in detail. For the channel under consideration with trapezoidal geometry (base width $b = 10$ m, side slope $1:m = 1:2$ and channel bed slope $S_0 = 0.25\%$), compute the water surface profile for a design discharge $Q = 25 \text{ m}^3/\text{s}$ and a Manning's roughness coefficient $n = 0.02857 \text{ s/m}^{1/3}$. The flow is subcritical with a uniform flow depth $y_0 = 2.25$ m. The proposed dam blocks the water at $x = 0$ m up to an elevation $y_p = 382$ m above mean sea level with a water depth at the dam of $y = 3.10$ m.

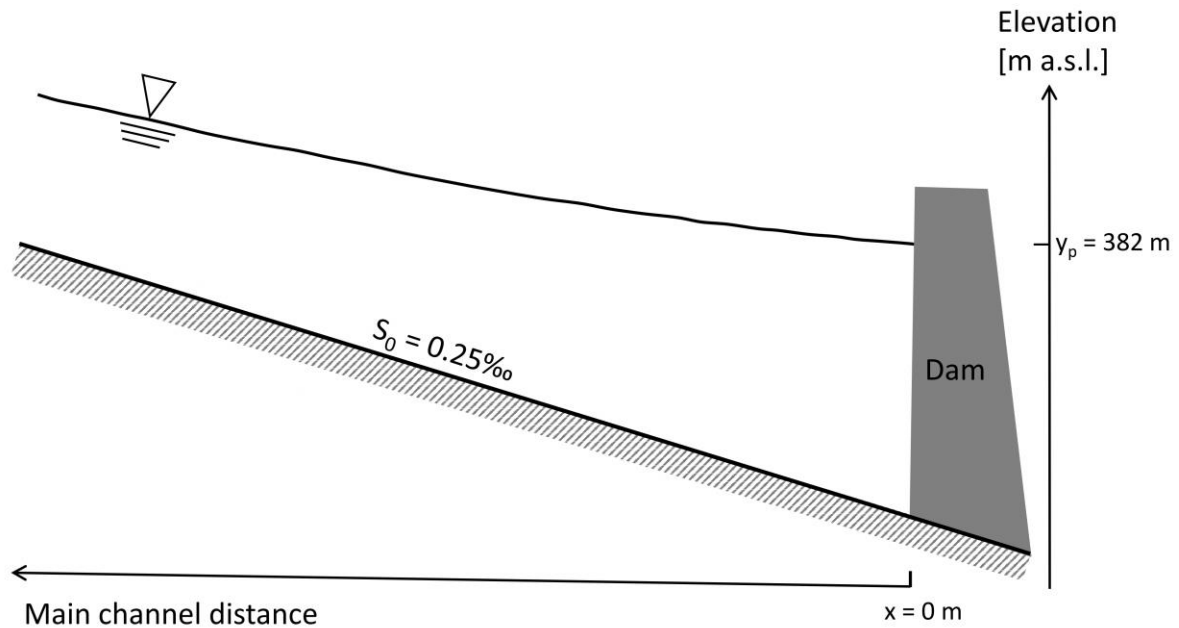


Figure 1a: Longitudinal-section view of the trapezoidal channel

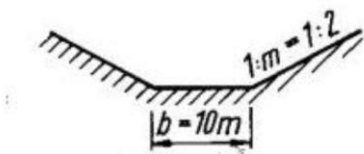


Figure 1b: Cross-section view of the trapezoidal channel

Determine the water surface profile using the standard step method.

Critical Depth Determination

Critical depth for a cross section will be determined if any of the following conditions are satisfied:

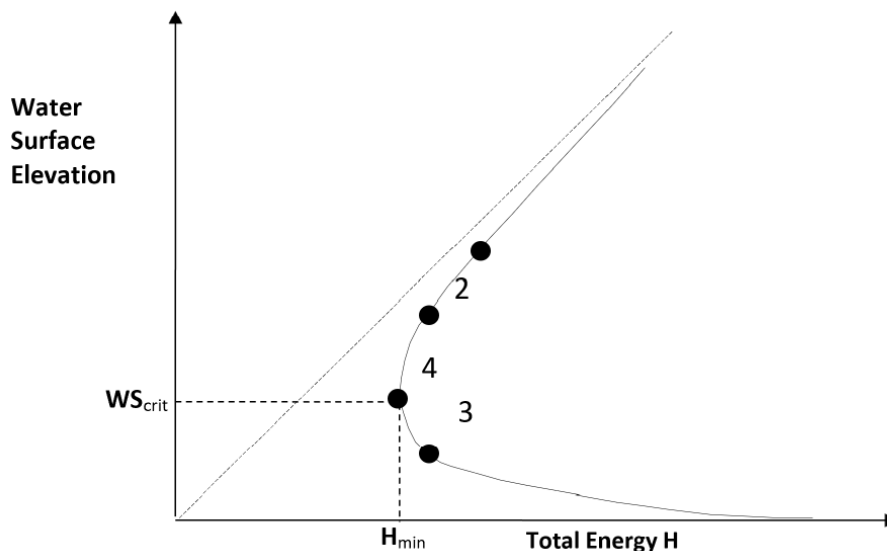
1. The supercritical flow regime has been specified.
2. The calculation of critical depth has been requested by the user.
3. This is an external boundary cross section and critical depth must be determined to ensure the user entered boundary condition is in the correct flow regime.
4. The Froude number check for a subcritical profile indicates that critical depth needs to be determined to verify the flow regime associated with the balanced elevation.
5. The program could not balance the energy equation within the specified tolerance before reaching the maximum number of iterations.

The total energy head for a cross section is defined by:

$$H = WS + \frac{aV^2}{2g} \quad (19)$$

Symbol	Description
H	total energy head
WS	water surface elevation
$\frac{aV^2}{2g}$	velocity head

The critical water surface elevation is the elevation for which the total energy head is a minimum (i.e., minimum specific energy for that cross section for the given flow). The critical elevation is determined with an iterative procedure whereby values of WS are assumed and corresponding values of H are determined with Equation (19) until a minimum value for H is reached.



The HEC-RAS program has two methods for calculating critical depth: a "parabolic" method and a "secant" method. The parabolic method is computationally faster, but it is only able to locate a single minimum energy. For most cross sections there will only be one minimum on the total energy curve, therefore the parabolic method has been set as the default method (the default method can be changed from the user interface). If the parabolic method is tried and it does not converge, then the program will automatically try the secant method. In certain situations, it is possible to have more than one minimum on the total energy curve. Multiple minimums are often associated with cross sections that have breaks in the total energy curve. These breaks can occur due to very wide and flat overbanks, as well as cross sections with levees and ineffective flow areas. When the parabolic method is used on a cross section that has multiple minimums on the total energy curve, the method will converge on the first minimum that it locates. This approach can lead to incorrect estimates of critical depth. If the user thinks that the program has incorrectly located critical depth, then the secant method should be selected and the model should be re-simulated. The "parabolic" method involves determining values of H for three values of WS that are spaced at equal ΔWS

intervals. The WS corresponding to the minimum value for H, defined by a parabola passing through the three points on the H versus WS plane, is used as the basis for the next assumption of a value for WS. It is presumed that critical depth has been obtained when there is less than a 0.003 m (0.01 ft) change in water depth from one iteration to the next and provided the energy head has not either decreased or increased by more than 0.003 m (0.01 feet). The "secant" method first creates a table of water surface versus energy by slicing the cross section into 30 intervals. If the maximum height of the cross section (highest point to lowest point) is less than 1.5 times the maximum height of the main channel (from the highest main channel bank station to the invert), then the program slices the entire cross section into 30 equal intervals. If this is not the case, the program uses 25 equal intervals from the invert to the highest main channel bank station, and then 5 equal intervals from the main channel to the top of the cross section. The program then searches this table for the location of local minimums. When a point in the table is encountered such that the energy for the water surface immediately above and immediately below are greater than the energy for the given water surface, then the general location of a local minimum has been found. The program will then search for the local minimum by using the secant slope projection method. The program will iterate for the local minimum either thirty times or until the critical depth has been bounded by the critical error tolerance. After the local minimum has been determined more precisely, the program will continue searching the table to see if there are any other local minimums.

The program can locate up to three local minimums in the energy curve. If more than one local minimum is found, the program sets critical depth equal to the one with the minimum energy. If this local minimum is due to a break in the energy curve caused by overtopping a levee or an ineffective flow area, then the program will select the next lowest minimum on the energy curve. If all of the local minimums are occurring at breaks in the energy curve (caused by levees and ineffective flow areas), then the program will set critical depth to the one with the lowest energy. If no local minimums are found, then the program will use the water surface elevation with the least energy. If the critical depth that is found is at the top of the cross section, then this is probably not a real critical depth. Therefore, the program will double the height of the cross section and try again. Doubling the height of the cross section is accomplished by extending vertical walls at the first and last points of the section. The height of the cross section can be doubled five times before the program will quit searching.

Recall:

$$\frac{Q^2}{g} = \frac{A^3}{T}$$

Where Q is the discharge, g is acceleration due to gravity, A is the cross-section area and T is the top width of the water surface.

$$Fr = \frac{V}{\left(\frac{gA}{T}\right)^{1/2}}$$

Where Fr is the Froude number and V is the flow velocity.

Applications of the Momentum Equation

Whenever the water surface passes through critical depth, the energy equation is not considered to be applicable. The energy equation is only applicable to gradually varied flow

situations, and the transition from subcritical to supercritical or supercritical to subcritical is a rapidly varying flow situation. There are several instances when the transition from subcritical to supercritical and supercritical to subcritical flow can occur. These include significant changes in channel slope, bridge constrictions, drop structures and weirs, and stream junctions. In some of these instances empirical equations can be used (such as at drop structures and weirs), while at others it is necessary to apply the momentum equation in order to obtain an answer.

Within HEC-RAS, the momentum equation can be applied for the following specific problems: the occurrence of a hydraulic jump; low flow hydraulics at bridges; and stream junctions. In order to understand how the momentum equation is being used to solve each of the three problems, a derivation of the momentum equation is shown here. The application of the momentum equation to hydraulic jumps and stream junctions is discussed in detail in the HEC-RAS Hydraulic Reference Manual. Detailed discussions on applying the momentum equation to bridges can be found in the section on "Modeling Bridges" in the HEC-RAS Hydraulic Reference Manual.

The momentum equation is derived from Newton's second law of motion:

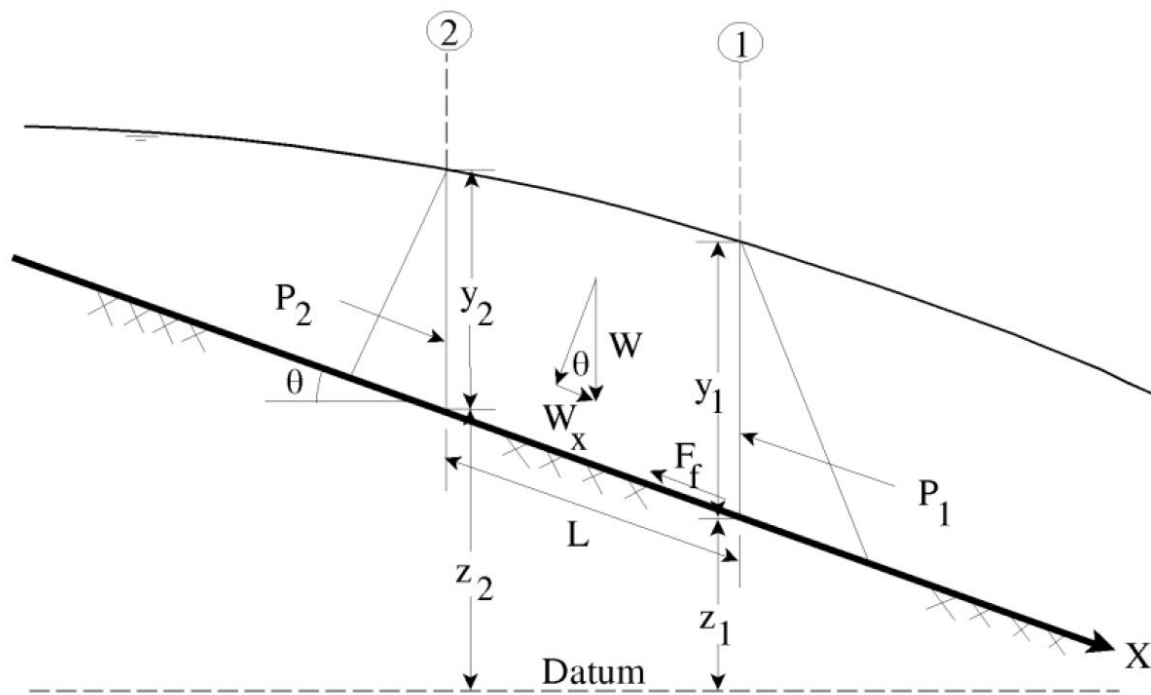
Force = Mass x Acceleration (change in momentum)

$$\sum F_x = ma \quad (20)$$

Applying Newton's second law of motion to a body of water enclosed by two cross sections at locations 1 and 2 (see Figure below), the following expression for the change in momentum over a unit time can be written:

$$P_2 - P_1 + W_x - F_f = Q\rho\Delta V_x \quad (21)$$

Symbol	Description
P	Hydrologic pressure force at locations 1 and 2.
W_x	Force due to the weight of water in the X direction.
F_f	Force due to external friction losses from 2 and 1
W	Discharge
ρ	Density of water
ΔV_x	Change on velocity from 2 to 1, in the X direction.



Hydrostatic Pressure Forces:

The force in the X direction due to hydrostatic pressure is:

$$P = \gamma A \bar{Y} \cos \theta \quad (22)$$

The assumption of a hydrostatic pressure distribution is only valid for slopes less than 1:10. The $\cos \theta$ for a slope of 1:10 (approximately 6 degrees) is equal to 0.995. Because the slope of ordinary channels is far less than 1:10, the $\cos \theta$ correction for depth can be set equal to 1.0 (Chow, 1959). Therefore, the equations for the hydrostatic pressure force at sections 1 and 2 are as follows:

$$P_1 = \gamma A_1 \bar{Y}_1 \quad (23)$$

$$P_2 = \gamma A_2 \bar{Y}_2 \quad (24)$$

Symbol	Description
γ	Unit weight of water
A_1	Wetted area of the cross section at locations 1 and 2

Symbol	Description
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$\bar{Y}_1$ Depth measured from water surface to the centroid of the cross sectional area at locations 1 and 2.

Weight of Water Force:

Weight of water = (unit weight of water) x (volume of water)

$$W = \gamma \left(\frac{A_1 + A_2}{2} \right) L \quad (25)$$

$$W_x = W \times \sin\theta \quad (26)$$

$$\sin\theta = \frac{z_2 - z_1}{L} = S_0 \quad (27)$$

$$W_x = \gamma \left(\frac{A_1 + A_2}{L} \right) L S_0 \quad (28)$$

Force of External Friction:

$$F_f = \tau \bar{P} L \quad (29)$$

Symbol	Description
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τ Shear stress

$\bar{P}$ Average wetted perimeter between sections 1 and 2

$$\tau = \gamma \bar{R} \bar{S}_f \quad (30)$$

Symbol	Description
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$\bar{R}$ Average hydraulic radius ($R = A/P$)

$\bar{S}_f$ Slope of the energy grade line (friction slope)

$$F_f = \gamma \frac{\bar{A}}{\bar{P}} \bar{S}_f \bar{P} L \quad (31)$$

$$F_f = \gamma \frac{A_1 + A_2}{2} \bar{S}_f L \quad (32)$$

Mass times Acceleration:

$$ma = Q \rho \Delta V_x \quad (33)$$

$$\rho = \frac{\gamma}{g} \text{ and } \Delta V_x = (\beta_1 V_1 - \beta_2 V_2)$$

$$ma = \frac{Q\gamma}{g}(\beta_1 V_1 - \beta_2 V_2) \quad (34)$$

Symbol	Description
β	momentum coefficient that accounts for a varying velocity distribution in irregular channel

Substituting Back into Equation (21), and assuming Q can vary from 2 to 1:

$$\gamma A_2 \bar{Y}_2 - \gamma A_1 \bar{Y}_1 + \gamma \left(\frac{A_1 + A_2}{2} \right) LS_0 - \gamma \left(\frac{A_1 + A_2}{2} \right) L\bar{S}_f = \frac{Q_1 \gamma}{g} \beta_1 V_1 - \frac{Q_2 \gamma}{g} \beta_2 V_2 \quad (35)$$

$$\frac{Q_2 \beta_2 V_2}{g} + A_2 \bar{Y}_2 + \left(\frac{A_1 + A_2}{2} \right) LS_0 - \left(\frac{A_1 + A_2}{2} \right) L\bar{S}_f = \frac{Q_1 \beta_1 V_1}{g} + A_1 \bar{Y}_1 \quad (36)$$

$$\frac{Q_2^2 \beta_2}{g A_2} + A_2 \bar{Y}_2 + \left(\frac{A_1 + A_2}{2} \right) LS_0 - \left(\frac{A_1 + A_2}{2} \right) L\bar{S}_f = \frac{Q_1 \beta_1}{g A_2} + A_1 \bar{Y}_1 \quad (37)$$

Equation (37) is the functional form of the momentum equation that is used in HEC-RAS. All applications of the momentum equation within HEC-RAS are derived from Equation 37.

1D Steady Flow HEC-RAS Program Limitations

The following assumptions are implicit in the analytical expressions used in the current version of the HEC-RAS program:

1. Flow is steady.
2. Flow is gradually varied. (Except at hydraulic structures such as: bridges; culverts; and weirs. At these locations, where the flow can be rapidly varied, the momentum equation or other empirical equations are used.)
3. Flow is one dimensional (i.e., velocity components in directions other than the direction of flow are not accounted for).
4. River channels have "small" slopes, say less than 1:10.

Flow is assumed to be steady because time dependent terms are not included in the energy equation (Equation (1)). Flow is assumed to be gradually varied because Equation (1) is based on the premise that a hydrostatic pressure distribution exists at each cross section. At locations where the flow is rapidly varied, the program switches to the momentum equation or other empirical equations. Flow is assumed to be one-dimensional because Equation (19) is based on the premise that the total energy head is the same for all points in a cross section. The limit on slope as being less than 1:10 is based on the fact that the true derivation of the energy equation computes the vertical pressure head as:

$$H_p = d \cos \theta$$

Symbol	Description
H_p	vertical pressure head
d	depth of the water measured perpendicular to the channel bottom.
θ	the channel bottom slope expressed in degrees.

For a channel bottom slope of 1:10 (5.71 degrees) or less, the $\cos(\theta)$ is 0.995. So instead of using $d \cos(\theta)$, the vertical pressure head is approximated as d and is used as the vertical depth of water. As you can see for a slope of 1:10 or less, this is a very small error in estimating the vertical depth (.5 %).

If HEC-RAS is used on steeper slopes, you must be aware of the error in the depth computation introduced by the magnitude of the slope. Below is a table of slopes and the $\cos(\theta)$:

Slope	Degrees	Cos (θ)
1:10	5.71	0.995
2:10	11.31	0.981
3:10	16.70	0.958
4:10	21.80	0.929
5:10	26.57	0.894

If you use HEC-RAS to perform the computations on slopes steeper than 1:10, you would need to divide the computed depth of water by the $\cos(\theta)$ in order to get the correct depth of water. Also, be aware that very steep slopes can introduce air entrainment into the flow, as well as other possible factors that may not be taken into account within HEC-RAS.

Air Entrainment in High Velocity Streams

For channels that have high flow velocity, the water surface may be slightly higher than otherwise expected due to the entrainment of air. While air entrainment is not important for most rivers, it can be significant for highly supercritical flows (Froude numbers greater than 1.6). HEC-RAS now takes this into account with the following two equations (EM 1110-2-1601, plate B-50)

For Froude numbers less than or equal to 1.6

$$D_a = 0.906 D (e)^{0.061 F} \quad (38)$$

For Froude numbers greater than 1.6

$$D_a = 0.620 D (e)^{0.1051 F} \quad (39)$$

Where: D_a = water depth with air entrainment

D = water depth without air entrainment

e = numerical constant, equal to 2.718282

F = Froude number

A water surface with air entrainment is computed and displayed separately in the HEC-RAS tabular output. In order to display the water surface with air entrainment, the user must create their own profile table and include the variable "WS Air Entr." within that table. This variable is not automatically displayed in any of the standard HEC-RAS tables.

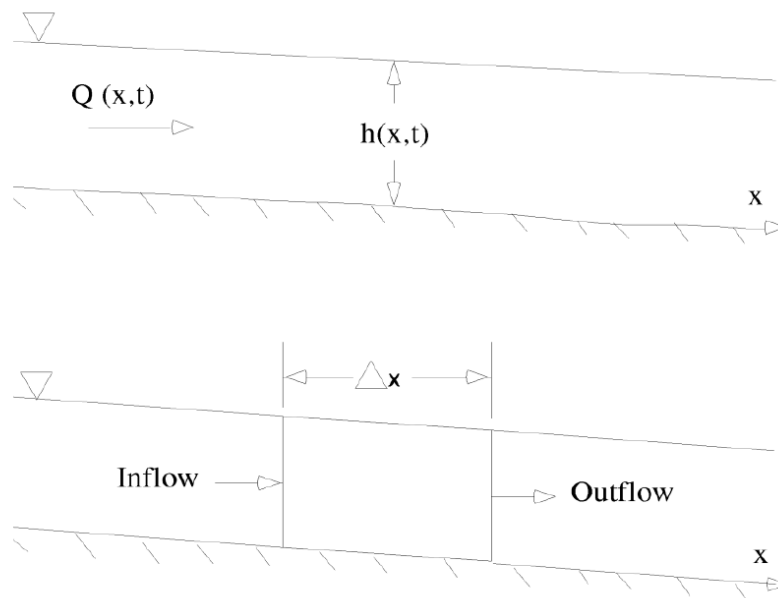
3 THEORETICAL BASIS FOR UNSTEADY ONE-DIMENSIONAL HYDRODYNAMIC CALCULATIONS

1D Unsteady Flow Hydrodynamics

The physical laws which govern the flow of water in a stream are: (1) the principle of conservation of mass (continuity), and (2) the principle of conservation of momentum. These laws are expressed mathematically in the form of partial differential equations, which will hereafter be referred to as the continuity and momentum equations. The derivations of these equations are presented in this chapter based on a paper by James A. Liggett from the book *Unsteady Flow in Open Channels* (Mahmmod and Yevjevich, 1975).

Continuity Equation

Consider the elementary control volume shown in the figure below. In this figure, distance x is measured along the channel, as shown. At the midpoint of the control volume the flow and total flow area are denoted $Q(x,t)$ and A_T , respectively. The total flow area is the sum of active area A and off channel storage area S .



Conservation of mass for a control volume states that the net rate of flow into the volume be equal to the rate of change of storage inside the volume. The rate of inflow to the control volume may be written as:

$$Q - \frac{\partial Q}{\partial x} \frac{\Delta x}{2} \quad (40)$$

the rate of outflow as:

$$Q + \frac{\partial Q}{\partial x} \frac{\Delta x}{2} \quad (41)$$

And the rate of change in storage as:

$$\frac{\partial A_T}{\partial t} \Delta x \quad (42)$$

Assuming that Δx is small, the change in mass in the control volume is equal to:

$$\rho \frac{\partial A_T}{\partial t} \Delta x = \rho \left[\left(Q - \frac{\partial Q}{\partial x} \frac{\Delta x}{2} \right) - \left(Q + \frac{\partial Q}{\partial x} \frac{\Delta x}{2} \right) + Q_l \right] \quad (43)$$

Symbol	Description
Q_l	is the lateral flow entering the control volume and ρ is the fluid density.

Simplifying and dividing through by ρ yields the final form of the continuity equation:

$$\frac{\partial A_T}{\partial t} + \frac{\partial Q}{\partial x} = q_l \quad (44)$$

Symbol	Description
q_l	lateral inflow per unit length

Momentum Equation

Conservation of momentum is expressed by Newton's second law as:

$$\sum F_x = \frac{d\vec{M}}{dt} \quad (45)$$

Conservation of momentum for a control volume states that *the net rate of momentum entering the volume (momentum flux) plus the sum of all external forces acting on the volume be equal to the rate of accumulation of momentum*. This is a vector equation applied in the x-direction. The momentum flux (M) is the fluid mass times the velocity vector in the direction of flow. Three forces will be considered: (1) pressure, (2) gravity and (3) boundary drag, or friction force.

Pressure forces: The Figure below illustrates the general case of an irregular cross section. The pressure distribution is assumed to be hydrostatic (pressure varies linearly with depth) and the total pressure force is the integral of the pressure-area product over the cross section. After Shames (1962), the pressure force at any point may be written as:

$$F_p = \int_0^h \rho g(h - y)T(y)dy \quad (46)$$

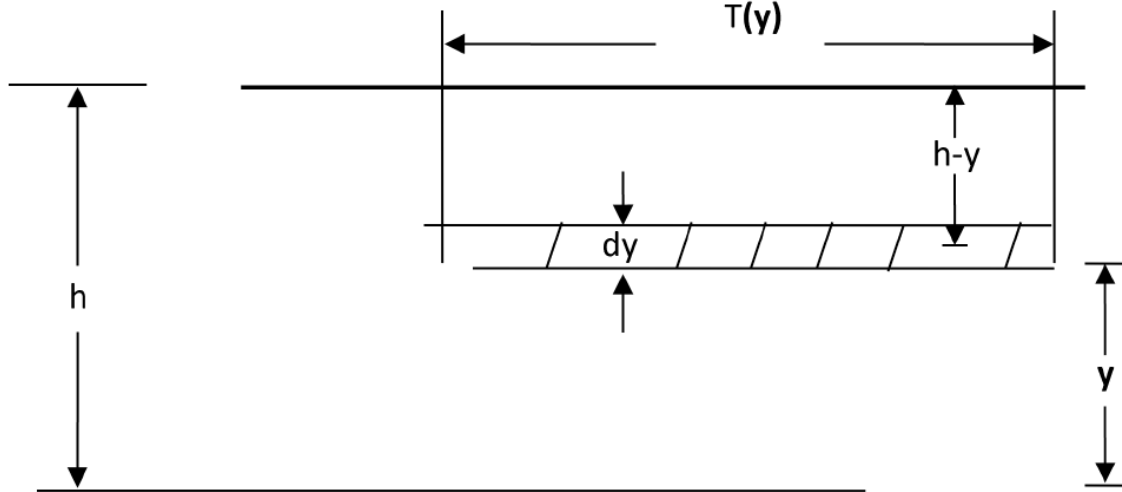
Symbol	Description
h	depth
y	distance above the channel invert
$T(y)$	width function which relates the cross section width to the distance above the channel invert

If F_p is the pressure force in the x-direction at the midpoint of the control volume, the force at the upstream end of the control volume may be written as

$$F_p - \frac{\partial F_p}{\partial x} \frac{\Delta x}{2} \quad (47)$$

and at the downstream end as:

$$F_p + \frac{\partial F_p}{\partial x} \frac{\Delta x}{2} \quad (48)$$



The sum of the pressure forces for the control volume may therefore be written as:

$$F_{Pn} = \left| F_p - \frac{\partial F_p}{\partial x} \frac{\Delta x}{2} \right| - \left| F_p + \frac{\partial F_p}{\partial x} \frac{\Delta x}{2} \right| + F_B \quad (49)$$

Where F_{Pn} is the net pressure force for the control volume, and F_B is the force exerted by the banks in the x-direction on the fluid. This may be simplified to:

$$F_{Pn} = -\frac{\partial F_p}{\partial x} \Delta x + F_B \quad (50)$$

Differentiating (46) using Leibnitz's Rule and then substituting in (50) results in:

$$F_{Pn} = -\rho g \Delta x \left[\frac{\partial h}{\partial x} \int_0^h T(y) dy + \int_0^h (h-y) \frac{\partial T(y)}{\partial x} dy \right] + F_B \quad (51)$$

The first integral in (51) is the cross-sectional area, A. The second integral (multiplied by $-\rho g \Delta x$) is the pressure force exerted by the fluid on the banks, which is exactly equal in magnitude, but opposite in direction to F_B . Hence the net pressure force may be written as:

$$F_{Pn} = -\rho g A \frac{\partial h}{\partial x} \Delta x \quad (52)$$

Gravitational force: The force due to gravity on the fluid in the control volume in the x-direction is:

$$F_g = \rho g A \sin \theta \Delta x \quad (53)$$

Here θ is the angle that the channel invert makes with the horizontal. For natural rivers θ is small and $\sin\theta \approx \tan\theta = -\frac{dz_0}{dx}$, where z_0 is the invert elevation. Therefore, the gravitational force may be written as

$$F_g = -\rho g A \frac{\partial z_0}{\partial x} \Delta x \quad (54)$$

This force will be positive for negative bed slopes.

Boundary drag (friction force): Frictional forces between the channel and the fluid may be written as:

$$F_f = -\tau_0 P \Delta x \quad (55)$$

Where τ_0 is the average boundary shear stress (force/unit area) acting on the fluid boundaries, and P is the wetted perimeter. The negative sign indicates that, with flow in the positive x -direction, the force acts in the negative x -direction. From dimensional analysis τ_0 , may be expressed in terms of a drag coefficient, C_D , as follows:

$$\tau_0 = \rho C_D V^2 \quad (56)$$

The drag coefficient may be related to the Chezy coefficient, C , by the following:

$$C_D = \frac{g}{C^2} \quad (57)$$

Further, the Chezy equation may be written as:

$$V = C \sqrt{R S_f} \quad (58)$$

Substituting (56), (57), and (58) into (55), and simplifying, yields the following expression for the boundary drag force:

$$F_f = -\rho g A S_f \Delta x \quad (59)$$

Where S_f is the friction slope, which is positive for flow in the positive x -direction. The friction slope must be related to flow and stage. Traditionally, the Manning and Chezy friction equations have been used. Since the Manning equation is predominantly used in the United States, it is also used in HEC-RAS. The Manning equation is written as:

$$S_f = \frac{Q|Q|n^2}{2.208R^{4/3}A^2} \quad (60a) \text{ English units}$$

$$S_f = \frac{Q|Q|n^2}{1.0R^{4/3}A^2} \quad (60b) \text{ SI units}$$

Symbol	Description
R	hydraulic radius and n is the Manning friction coefficient

Momentum flux: With the three force terms defined, only the momentum flux remains. The flux entering the control volume may be written as:

$$\rho \left[QV - \frac{\partial QV}{\partial x} \frac{\Delta x}{2} \right] \quad (61)$$

and the flux leaving the volume may be written as:

$$\rho \left[QV + \frac{\partial QV}{\partial x} \frac{\Delta x}{2} \right] \quad (62)$$

Therefore, the net rate of momentum (momentum flux) entering the control volume is:

$$-\rho \frac{\partial(QV)}{\partial x} \Delta x \quad (63)$$

Since the momentum of the fluid in the control volume is $\rho Q \Delta x$, the rate of accumulation of momentum may be written as:

$$\frac{\partial}{\partial t}(\rho Q \Delta x) = \rho \Delta x \frac{\partial Q}{\partial t} \quad (64)$$

Restating the principle of conservation of momentum:

The net rate of momentum (momentum flux) entering the volume (63) plus the sum of all external forces acting on the volume [(52) + (54) + (59)] is equal to the rate of accumulation of momentum (64). Hence:

$$\rho \Delta x \frac{\partial Q}{\partial t} = -\rho \frac{\partial(QV)}{\partial x} \Delta x - \rho g A \frac{\partial h}{\partial x} \Delta x - \rho g A \frac{\partial z_0}{\partial x} \Delta x - \rho g A S_f \Delta x \quad (65)$$

The elevation of the water surface, z , is equal to $z_0 + h$. Therefore:

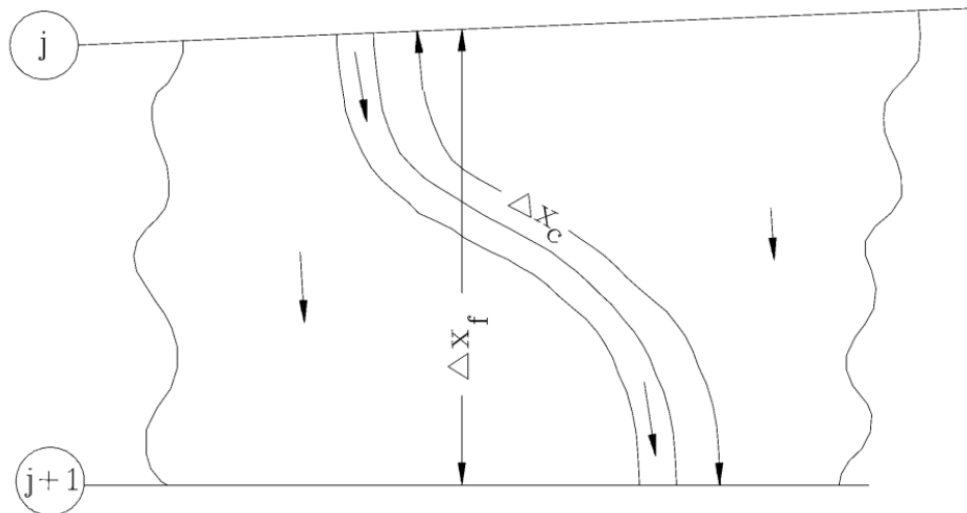
$$\frac{\partial z}{\partial x} = \frac{\partial h}{\partial x} + \frac{\partial z_0}{\partial x} \quad (66)$$

Where $\partial z / \partial x$ is the water surface slope. Substituting (66) into (65), dividing through by $\rho \Delta x$ and moving all terms to the left yields the final form of the momentum equation:

$$\frac{\partial Q}{\partial t} + \frac{\partial(QV)}{\partial x} + gA \left(\frac{\partial z}{\partial x} + S_f \right) = 0 \quad (67)$$

Application of the 1D Unsteady Flow Equations within HEC-RAS

The figure below illustrates the two-dimensional characteristics of the interaction between the channel and floodplain flows. When the river is rising water moves laterally away from the channel, inundating the floodplain and filling available storage areas. As the depth increases, the floodplain begins to convey water downstream generally along a shorter path than that of the main channel. When the river stage is falling, water moves toward the channel from the overbank supplementing the flow in the main channel.



Because the primary direction of flow is oriented along the channel, this two-dimensional flow field can often be accurately approximated by a one-dimensional representation. Off-channel ponding areas can be modelled with storage areas that exchange water with the channel. Flow in the overbank can be approximated as flow through a separate channel. This channel/floodplain problem has been addressed in many different ways. A common approach is to ignore overbank conveyance entirely, assuming that the overbank is used only for storage. This assumption may be suitable for large streams such as the Mississippi River where the channel is confined by levees and the remaining floodplain is either heavily vegetated or an off-channel storage area. Fread (1976) and Smith (1978) approached this problem by dividing the system into two separate channels and writing continuity and momentum equations for each channel. To simplify the problem, they assumed a horizontal water surface at each cross section normal to the direction of flow; such that the exchange of momentum between the channel and the floodplain was negligible and that the discharge was distributed according to conveyance, i.e.:

$$Q_c = \phi Q \quad (68)$$

Symbol	Description
Q_c	flow in channel
Q	total flow
ϕ	$\frac{K_c}{K_c + K_f}$
K_c	conveyance in the channel
K_f	conveyance in the floodplain

With these assumptions, the one-dimensional equations of motion can be combined into a single set:

$$\frac{\partial A}{\partial t} + \frac{\partial(\Phi Q)}{\partial x_c} + \frac{\partial[(1 - \Phi)Q]}{\partial x_f} = 0 \quad (69)$$

$$\frac{\partial Q}{\partial t} + \frac{\partial \left(\frac{\Phi^2 Q^2}{A_c} \right)}{\partial x_c} + \frac{\partial \left(\frac{(1-\Phi^2) Q^2}{A_f} \right)}{\partial x_f} + g A_c \left[\frac{\partial Z}{\partial x_c} + S_{fc} \right] + g A_f \left[\frac{\partial z}{\partial x_f} + S_{ff} \right] = 0 \quad (70)$$

in which the subscripts c and f refer to the channel and floodplain, respectively. These equations were approximated using implicit finite differences, and solved numerically using the Newton-Raphson iteration technique. The model was successful and produced the desired effects in test problems. Numerical oscillations, however, can occur when the flow at one node, bounding a finite difference cell, is within banks and the flow at the other node is not. Expanding on the earlier work of Fread and Smith, Barkau (1982) manipulated the finite difference equations for the channel and floodplain and defined a new set of equations that were computationally more convenient. Using a velocity distribution factor, he combined the convective terms. Further, by defining an equivalent flow path, Barkau replaced the friction slope terms with an equivalent force. The equations derived by Barkau are the basis for the unsteady flow solution within the HEC-RAS software. These equations were derived above. The numerical solution of these equations is described in detail in the HEC-RAS Hydraulic Reference Manual.

APPENDIX

STEADY VS UNSTEADY FLOW WITH HEC-RAS (*discussion by Victor Ponce*)

The U.S. Army Corps of Engineers HEC-RAS model (since version 5.0) can perform the following five functions: (1) one-dimensional steady flow, (2) one-dimensional unsteady flow, (3) two-dimensional unsteady flow, (4) mobile-bed fluvial hydraulics, and (5) water temperature and water quality modelling. Here we compare the one-dimensional steady and unsteady flow modules.

The one-dimensional steady flow component uses the standard step method for the solution of steady gradually varied flow.¹ The one-dimensional unsteady flow component uses a numerical solution of the equations governing gradually varied unsteady flow in open channels.

When should unsteady flow be used?

This question is of considerable practical interest, since unsteady flow is significantly more complex and requires more data than steady flow. However, the answer is not straightforward, requiring some elaboration.

Steady vs unsteady flow

Under steady flow, the user inputs as boundary conditions a discharge upstream and a stage downstream. The model proceeds to calculate stages throughout the interior points, keeping the discharge constant in space. Under unsteady flow, the user inputs a discharge hydrograph at the upstream boundary and a discharge-stage rating at the downstream boundary. The model calculates discharges and stages throughout the interior points.

Under steady flow, the calculated discharge-stage ratings are unique, i.e., kinematic. On the other hand, under unsteady flow, the model calculates dynamic looped discharge-stage ratings according to the variabilities of the flow. Therefore, the specification of a unique discharge-stage rating at the downstream boundary contradicts the solution at that boundary.² In essence, the model cannot be kinematic at the downstream boundary and dynamic everywhere else.

A way out of the predicament is: (1) to artificially move the downstream boundary further downstream, (2) to specify the unique discharge-stage rating at the artificial downstream boundary, and (3) to let the model calculate the looped ratings at the interior points, including the point where the physical downstream boundary is located.³ Despite its apparent artificiality, this procedure works well, circumventing the need to know the discharge-stage rating at the physical downstream boundary before it is calculated.

Kinematic vs dynamic waves

The decision to use unsteady flow instead of steady flow will depend on whether the wave to be modelled is construed as either kinematic or dynamic.

If the wave is kinematic: (1) the discharge will not vary in space; (2) the discharge-stage ratings will be unique; and (3) the downstream boundary may be specified as unique. In this case, the solutions of steady and unsteady flow are essentially the same; therefore, the unsteady flow calculation is not necessary.

If the wave is dynamic: (1) the discharge will vary in space, attenuating as it moves downstream, albeit only for Froude numbers less than 2, which is usually the case; (2) the calculated discharge-stage ratings will not be unique; and (3) for better accuracy, the

downstream boundary may be artificially moved downstream to allow for an unsteady looped rating to develop at the physical downstream boundary. In this case, the unsteady flow calculation is justified.

Use of unsteady flow in channel design

This situation begs the question of whether a certain flood wave can be construed as being either kinematic or dynamic. Or, alternatively, whether a dynamic wave should be used at all to determine stages in the design of channel improvement projects. On typical projects, of limited channel lengths, a kinematic wave, which keeps its discharge constant, is a better assumption than a dynamic wave, which attenuates its discharge. Indeed, the kinematic wave assumption assures that the channel will contain all waves types, kinematic, diffusion, and/or dynamic. Viewed in this light, the use of a dynamic wave for the calculation of stages in the design of channel improvements projects does not appear to be warranted.

In summary, the choice will always depend on the intended use. And the user must carefully select the most appropriate option. There will be cases where steady flow calculations are adequate, but there will be cases where unsteady flow calculations are absolutely necessary. In practice, the choice must be justified.

¹ Chow, V. T. (1959). Open-channel hydraulics. McGraw-Hill.

² Abbott, M. (1976). Computational hydraulics: A short pathology. Journal of Hydraulic Research, Vol. 14, No. 4.

³ Ponce, V. M. and A. Lugo. (2001). Modeling looped ratings in Muskingum-Cunge routing. ASCE Journal of Hydrologic Engineering, Vol. 6, No. 2, March/April, 119-124.

BASIC DATA REQUIREMENTS FOR HEC-RAS APPLICATIONS

This section describes the basic data requirements for performing the one-dimensional flow calculations within HEC-RAS. The basic data are defined and discussions of applicable ranges for parameters are provided. For information of data requirements for 2D modelling, please review the 2D Modelling User's manual. The main objective of the HEC-RAS program is quite simple - to compute water surface elevations at all locations of interest for either a given set of flow data (steady flow simulation), or by routing hydrographs through the system (unsteady flow simulation). The data needed to perform these computations are divided into the following categories: geometric data; steady flow data; unsteady flow data; sediment data; and water quality data. Geometric data are required for any of the analyses performed within HEC-RAS. The other data types are only required if you are going to do that specific type of analysis (i.e., steady flow data are required to perform a steady flow water surface profile computation). The current version of HEC-RAS can perform either steady or unsteady flow computations.

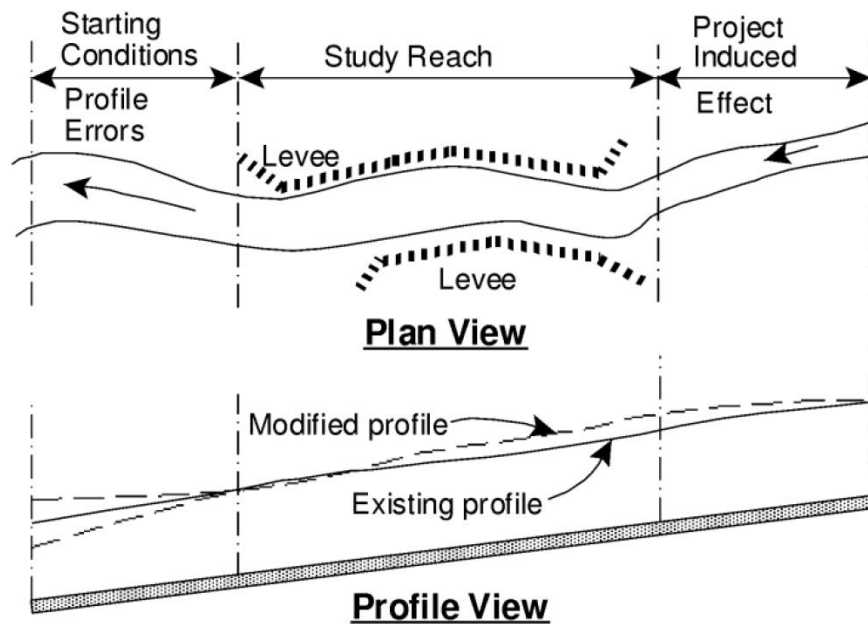
Geometric Data

Note

Before you begin to create the geometric data for an HEC-RAS hydraulic model, in general you will need to create a terrain model in HEC-RAS Mapper. A terrain model is not required for 1D modelling, as the user can enter all of the cross-sectional information, structure data, etc. directly into the HEC-RAS geometric data editor. However, to perform any 2D modelling or inundation mapping, a terrain model is required. Terrain models for HEC-RAS are created using HEC-RAS Mapper. Please review how to create a terrain model in the HEC-RAS Mapper User's manual. The basic geometric data consist of establishing the connectivity of the river system (River System Schematic); cross section data; reach lengths; energy loss coefficients (friction losses, contraction and expansion losses); stream junction information; storage areas; and 2D Flow Areas. Hydraulic structure data (bridges, culverts, spillways, weirs, etc...), which are also considered geometric data, will be described in other course materials.

Study Limit Determination

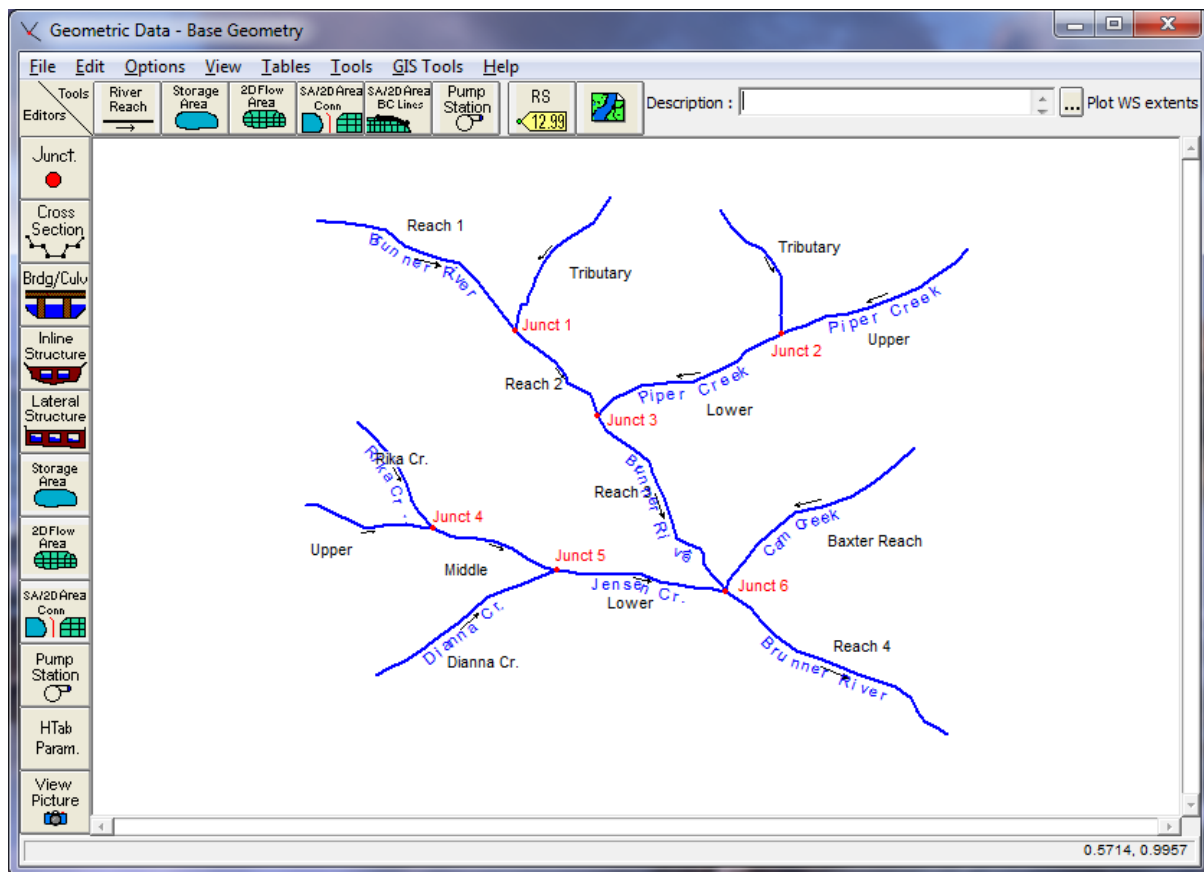
When performing a hydraulic study, it is normally necessary to gather data both upstream of and downstream of the study reach. Gathering additional data upstream is necessary in order to evaluate any upstream impacts due to construction alternatives that are being evaluated within the study reach (see Figure below). The limits for data collection upstream should be at a distance such that the increase in water surface profile resulting from a channel modification converges with the existing conditions profile. Additional data collection downstream of the study reach is necessary in order to prevent any user-defined boundary condition from affecting the results within the study reach. In general, the water surface at the downstream boundary of a model is not normally known. The user must estimate this water surface for each profile to be computed. A common practice is to use Manning's equation and compute normal depth as the starting water surface. The actual water surface may be higher or lower than normal depth. The use of normal depth will introduce an error in the water surface profile at the boundary. In general, for subcritical flow, the error at the boundary will diminish as the computations proceed upstream. In order to prevent any computed errors within the study reach, the unknown boundary condition should be placed far enough downstream such that the computed profile will converge to a consistent answer by the time the computations reach the downstream limit of the study.



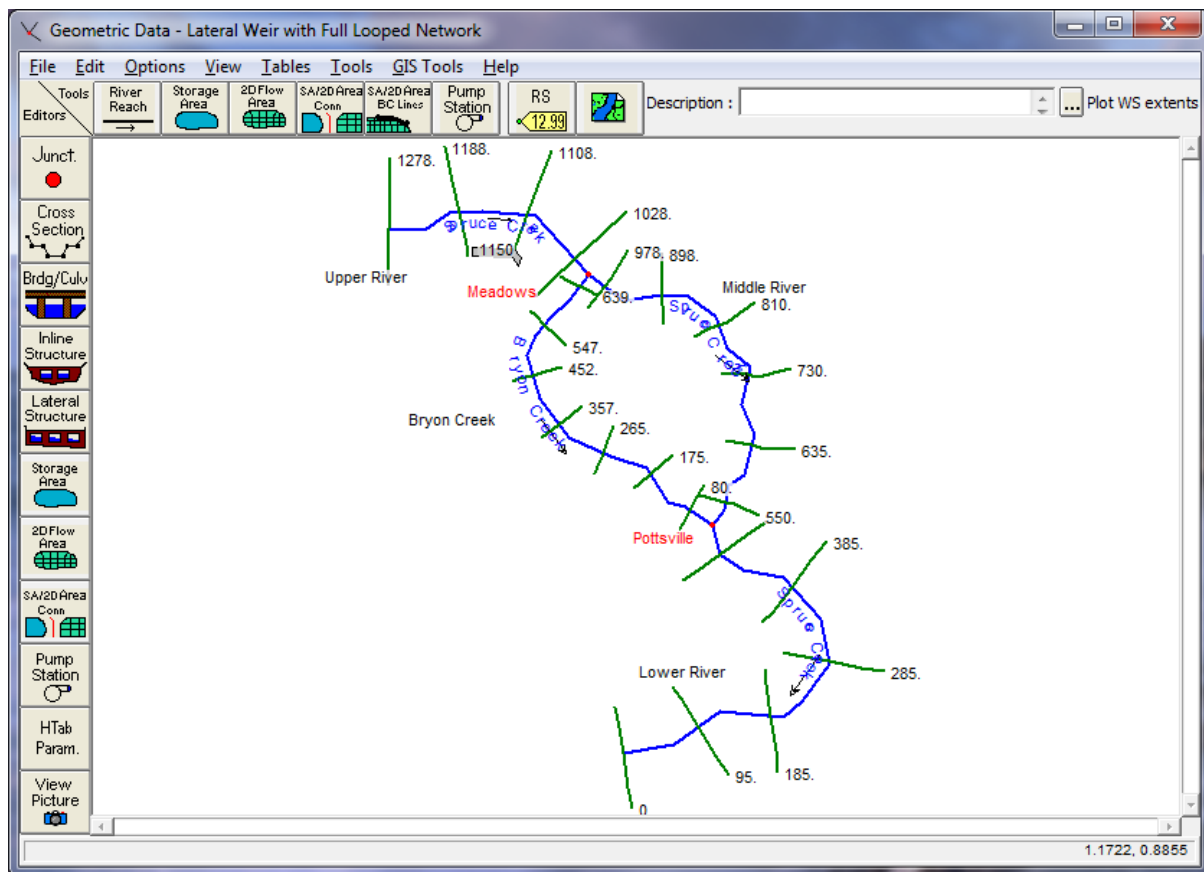
The River System Schematic

The river system schematic is required for any geometric data set within the HEC-RAS system. The schematic defines how the various river reaches, storage areas, and 2D flow areas are connected, as well as establishing a naming convention for referencing all the other data. The river system schematic is developed by drawing and connecting the various hydraulic elements of the system within the geometric data editor (see the HEC-RAS Mapper User's manual for how to layout the model schematic in a geospatial manner. See "Modeling Bridges" of the HEC-RAS User's Manual (<https://www.hec.usace.army.mil/confluence/rasdocs>) for details on how to develop the schematic from within the user interface). The user is required to develop the river system schematic before any other data can be entered.

Each river reach on the schematic is given a unique identifier. As other river reach data are entered, the data are referenced to a specific reach of the schematic. For example, each cross section must have a "River", "Reach" and "River Station" identifier. The river and reach identifiers define which reach the cross section lives in, while the river station identifier defines where that cross section is located within the reach, with respect to the other cross sections for that reach. The connectivity of reaches is very important for the model to understand how the computations should proceed from one reach to the next. The user is required to draw each reach from upstream to downstream, in what is considered to be the positive flow direction. The connecting of reaches is considered a junction. Junctions should only be established at locations where two or more streams come together or split apart. Junctions cannot be established with a single reach flowing into another single reach. These two reaches must be combined and defined as one reach. An example river system schematic is shown in the Figure below.



The example schematic shown in the Figure above is for a dendritic river system. Arrows are automatically drawn on the schematic in the assumed positive flow direction. Junctions (red circles) are automatically formed as reaches are connected. As shown, the user is required to provide a river and reach identifier for each reach, as well as an identifier for each junction. HEC-RAS has the ability to model river systems that range from a single reach model to complicated networks. A "network" model is where river reaches split apart and then come back together, forming looped systems. An example schematic of a looped stream network is shown in the Figure below.

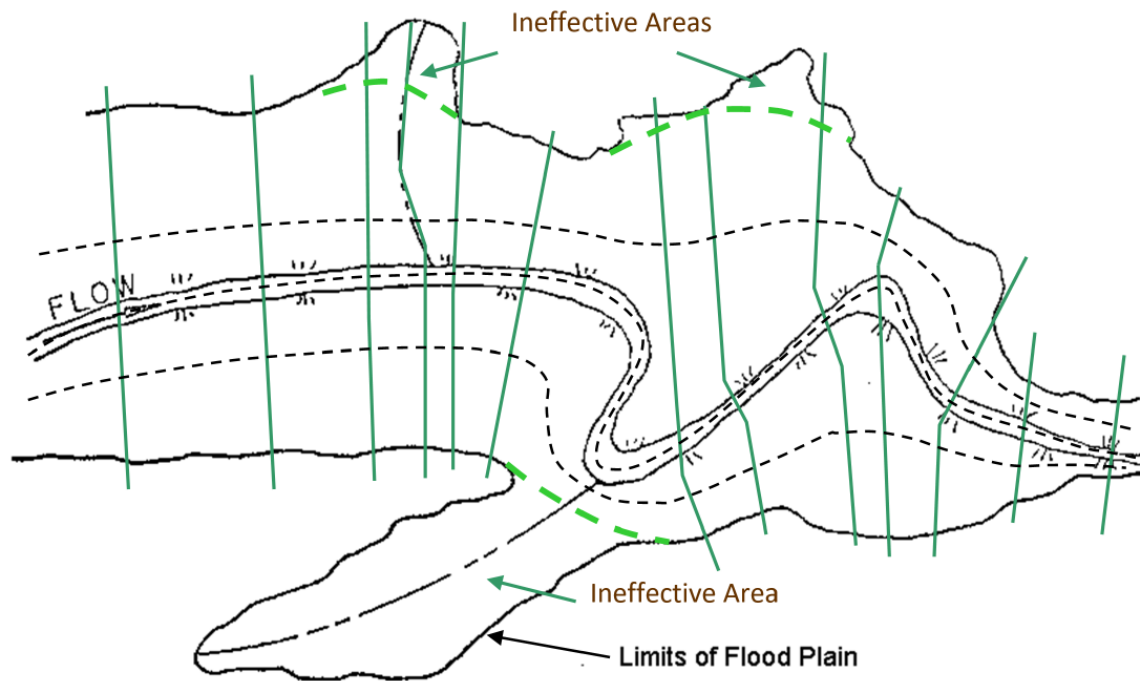


The river system schematic shown in the Figure above demonstrates the ability of HEC-RAS to model flow splits as well as flow combinations. The current version of the steady flow model within HEC-RAS does not determine the amount of flow going to each reach at a flow split, unless the user turns on the split flow optimization option.

Cross Section Geometry

Boundary geometry for the analysis of flow in natural streams is specified in terms of ground surface profiles (cross sections) and the measured distances between them (reach lengths). Cross sections are located at intervals along a stream to characterize the flow carrying capability of the stream and its adjacent floodplain. They should extend across the entire floodplain and should be perpendicular to the anticipated flow lines. Occasionally it is necessary to layout cross-sections in a curved or dogleg alignment to meet this requirement. Every effort should be made to obtain cross sections that accurately represent the stream and floodplain geometry.

An example of laying out cross sections is shown below in the figure below. The general approach to laying out cross sections is to ensure that the cross sections are perpendicular to the flow lines. This requires an estimation of what the flow lines will look like in the overbank areas away from the main channel. One option is to draw a stream centre line down the main channel along what is perceived to be the centre of mass of flow. The same thing should be done for the left overbank and the right overbank. The assumed flow paths for the channel and overbank areas are shown as dashed lines in the figure below. These lines will not only help in drawing the cross sections perpendicular to the flow lines, but they also represent the path for measuring the reach lengths between the cross sections.



Cross sections are required at representative locations throughout a stream reach and at locations where changes occur in discharge, slope, shape, or roughness, at locations where levees begin or end and at bridges or control structures such as weirs. Where abrupt changes occur, several cross sections should be used to describe the change regardless of the distance. Cross section spacing is also a function of stream size, slope, and the uniformity of cross section shape. In general, large uniform rivers of flat slope normally require the fewest number of cross sections per mile. The purpose of the study also affects spacing of cross sections. For instance, navigation studies on large relatively flat streams may require closely spaced (e.g., 60 metres) cross sections to analyse the effect of local conditions on low flow depths, whereas cross sections for sedimentation studies, to determine deposition in reservoirs, may be spaced at intervals on the order of thousands of feet. The choice of friction loss equation may also influence the spacing of cross sections. For instance, cross section spacing may be maximized when calculating an M1 profile (backwater profile) with the average friction slope equation or when the harmonic mean friction slope equation is used to compute M2 profiles (draw down profile). The HEC-RAS software provides the option to let the program select the averaging equation.

Each cross section in an HECRAS data set is identified by a River, Reach, and River Station label. The cross section is described by entering the station and elevation (X-Y data) from left to right, with respect to looking in the downstream direction. The River Station identifier may correspond to stationing along the channel, mile points, or any fictitious numbering system. The numbering system must be consistent, in that the program assumes that higher numbers are upstream and lower numbers are downstream.

Each data point in the cross section is given a station number corresponding to the horizontal distance from a starting point on the left. Up to 500 data points may be used to describe each cross section. Cross section data are traditionally defined looking in the downstream direction. The program considers the left side of the stream to have the lowest station numbers and the right side to have the highest. Cross section data are allowed to have negative stationing values. Stationing must be entered from left to right in increasing order.

However, more than one point can have the same stationing value. The left and right stations separating the main channel from the overbank areas must be specified on the cross-section data editor. End points of a cross section that are too low (below the computed water surface elevation) will automatically be extended vertically and a note indicating that the cross section had to be extended will show up in the output for that section.

The program adds additional wetted perimeter for any water that comes into contact with the extended walls.

Other data that are required for each cross section consist of: downstream reach lengths; roughness coefficients; and contraction and expansion coefficients. These data will be discussed in detail later in this chapter.

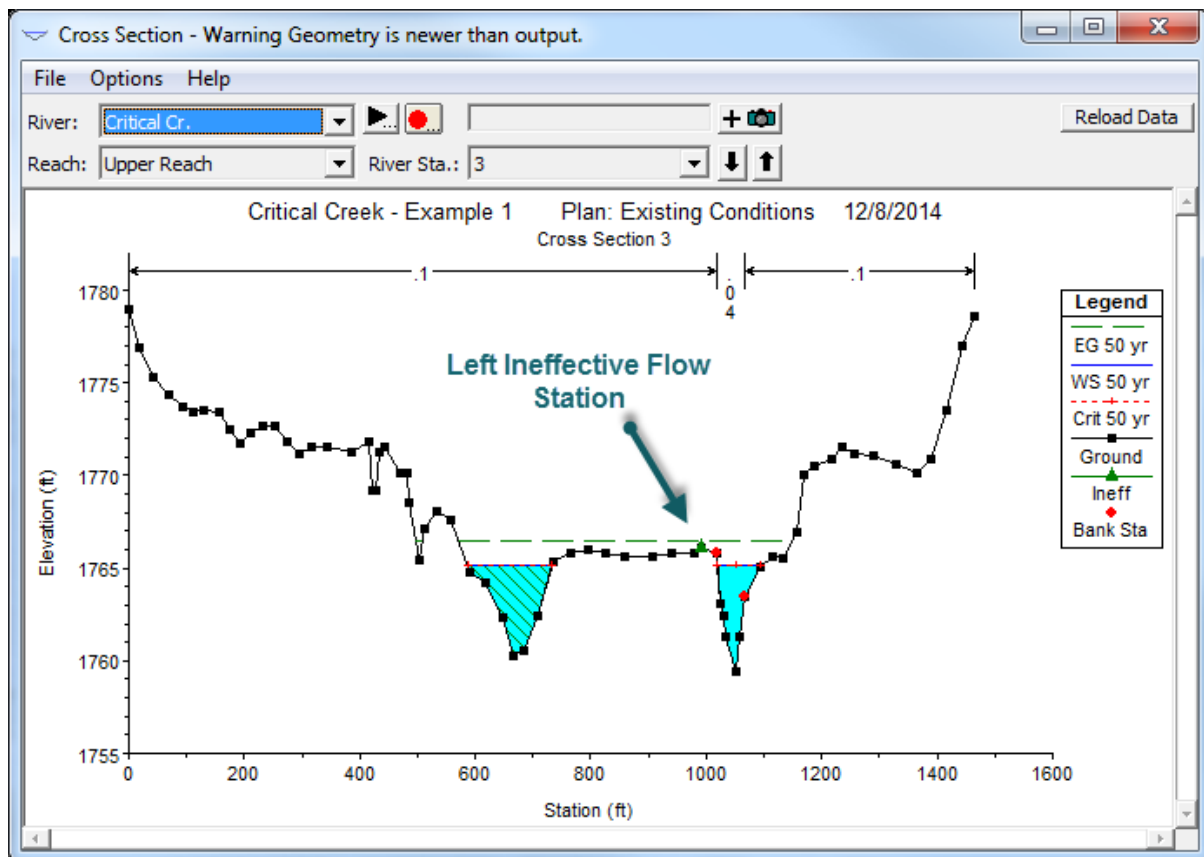
Numerous program options are available to allow the user to easily add or modify cross section data. For example, when the user wishes to repeat a surveyed cross section, an option is available from the interface to make a copy of any cross section. Once a cross section is copied, other options are available to allow the user to modify the horizontal and vertical dimensions of the repeated cross section data. For a detailed explanation on how to use these cross-section options, see "Modeling Culverts" of the HEC-RAS User's Manual and the Hydraulic Reference Manual (<https://www.hec.usace.army.mil/confluence/rasdocs>).

Optional Cross Section Properties

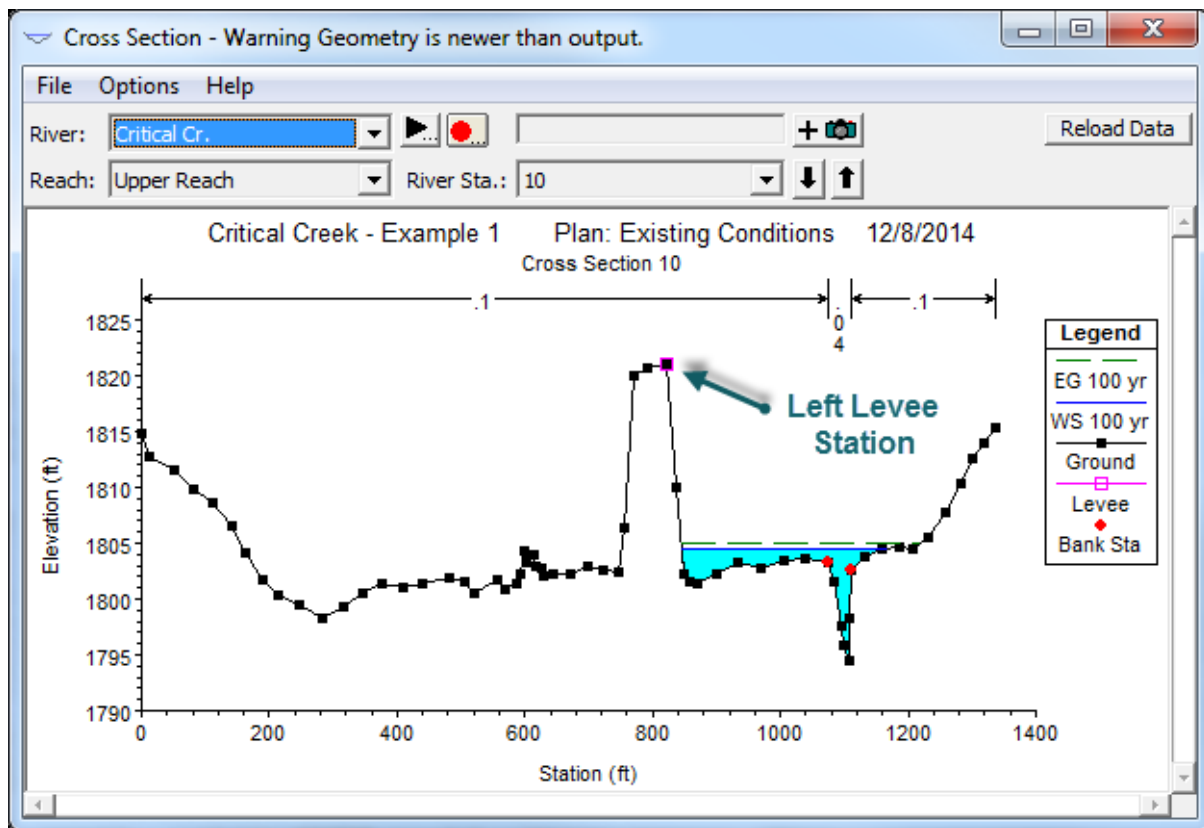
A series of program options are available to restrict flow to the effective flow areas of cross sections. Among these capabilities are options for: ineffective flow areas; levees; and blocked obstructions. All of these capabilities are available from the "Options" menu of the Cross Section Data editor. **Ineffective Flow Areas.** This option allows the user to define areas of the cross section that will contain water that is not actively being conveyed (ineffective flow). Ineffective flow areas are often used to describe portions of a cross section in which water will pond, but the velocity of that water, in the downstream direction, is close to zero. This water is included in the storage calculations and other wetted cross section parameters, but it is not included as part of the active flow area. When using ineffective flow areas, no additional wetted perimeter is added to the active flow area. An example of an ineffective flow area is shown in the figure below. The cross-hatched area on the left of the plot represents what is considered to be the ineffective flow.

Two alternatives are available for setting ineffective flow areas. The first option allows the user to define a left station and elevation and a right station and elevation (normal ineffective areas). When this option is used, and if the water surface is below the established ineffective elevations, the areas to the left of the left station and to the right of the right station are considered ineffective. Once the water surface goes above either of the established elevations, then that specific area is no longer considered ineffective.

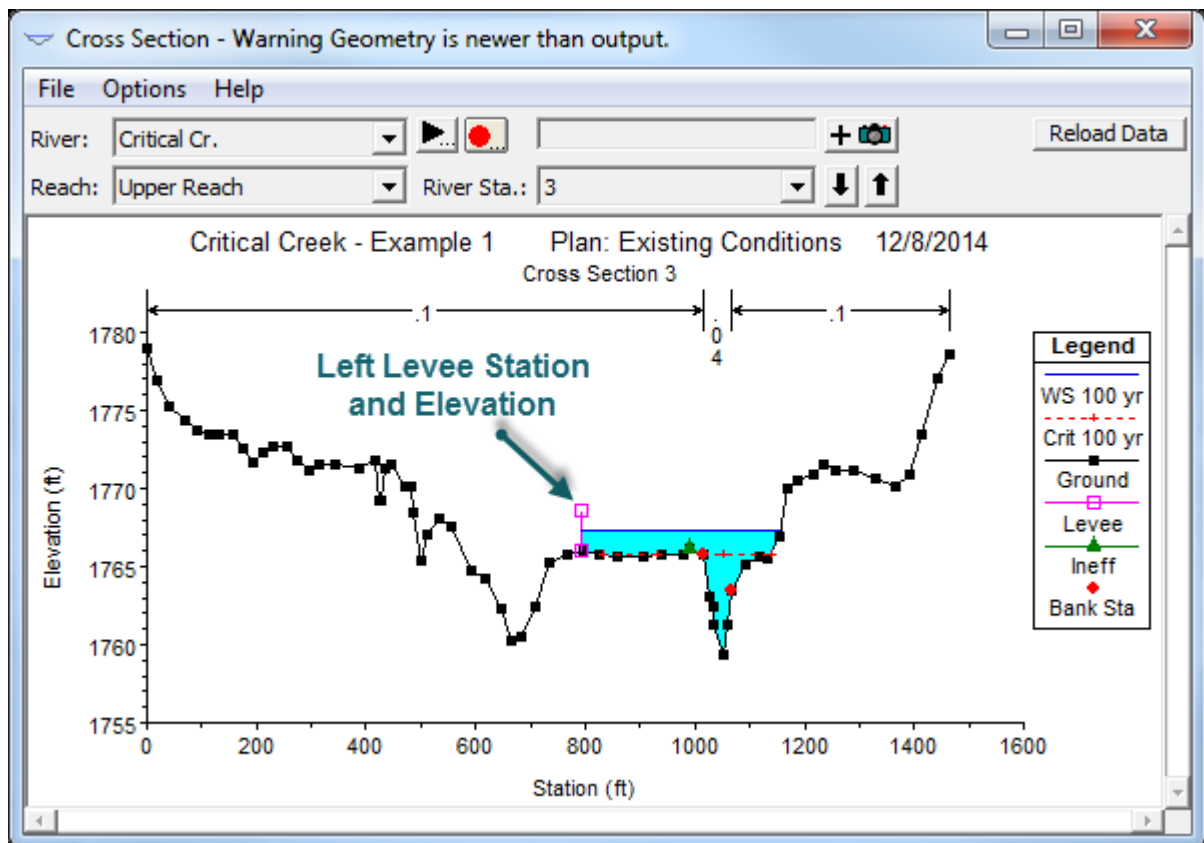
The second option allows for the establishment of blocked ineffective flow areas. Blocked ineffective flow areas require the user to enter an elevation, a left station, and a right station for each ineffective block. Up to ten blocked ineffective flow areas can be entered at each cross section. Once the water surface goes above the elevation of the blocked ineffective flow area, the blocked area is no longer considered ineffective.



Levees/stopbanks. This option allows the user to establish a left and/or right levee station and elevation on any cross section. When levees are established, no water can go to the left of the left levee station or to the right of the right levee station until either of the levee elevations are exceeded. Levee stations must be defined explicitly, or the program assumes that water can go anywhere within the cross section. An example of a cross section with a levee on the left side is shown in the figure below. In this example the levee station and elevation is associated with an existing point on the cross section.

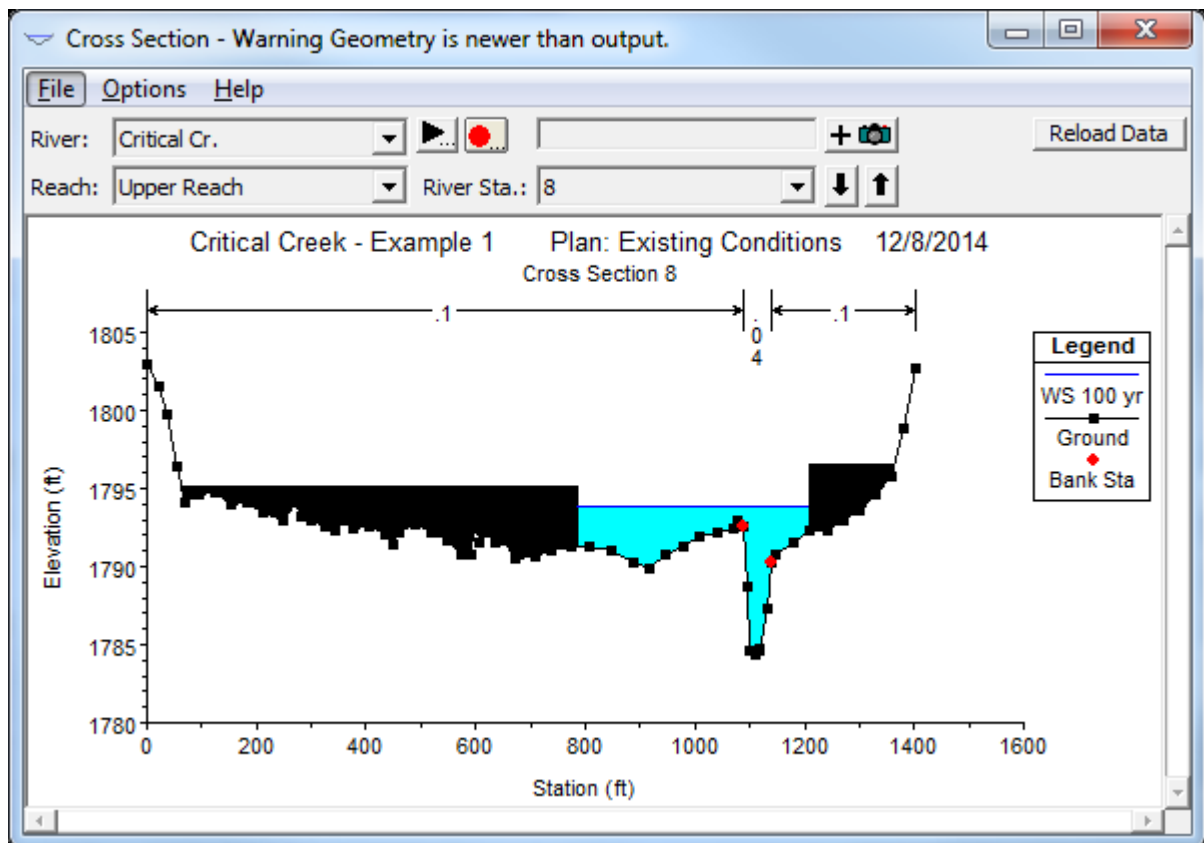


The user may want to add levees into a data set in order to see what effect a levee will have on the water surface. A simple way to do this is to set a levee station and elevation that is above the existing ground. If a levee elevation is placed above the existing geometry of the cross section, then a vertical wall is placed at that station up to the established levee height. Additional wetted perimeter is included when water comes into contact with the levee wall. An example of this is shown in the figure below.

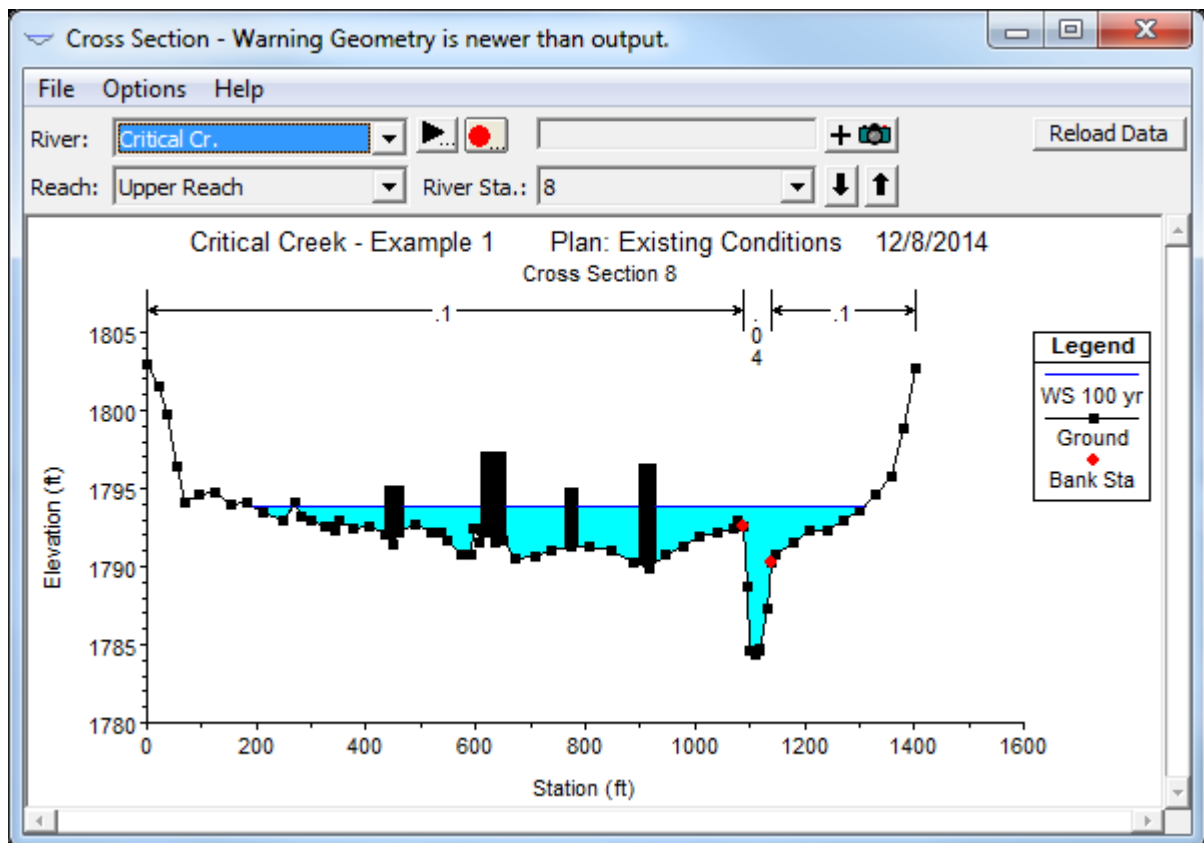


Obstructions. This option allows the user to define areas of the cross section that will be permanently blocked out. Obstructions decrease flow area and add wetted perimeter when the water comes in contact with the obstruction. A obstruction does not prevent water from going outside of the obstruction.

Two alternatives are available for entering obstructions. The first option allows the user to define a left station and elevation and a right station and elevation (normal obstructions). When this option is used, the area to the left of the left station and to the right of the right station will be completely blocked out. An example of this type of obstruction is shown in the figure below.



The second option, for obstructions, allows the user to enter up to 20 individual blocks (Multiple Blocks). With this option the user enters a left station, a right station, and an elevation for each of the blocks. An example of a cross section with multiple blocked obstructions is shown in the figure below.



Reach Lengths

The measured distances between cross sections are referred to as reach lengths. The reach lengths for the left overbank, right overbank and channel are specified on the cross section data editor. Channel reach lengths are typically measured along the thalweg. Overbank reach lengths should be measured along the anticipated path of the centre of mass of the overbank flow. Often, these three lengths will be of similar value. There are, however, conditions where they will differ significantly, such as at river bends, or where the channel meanders and the overbanks are straight. Where the distances between cross sections for channel and overbanks are different, a discharge weighted reach length is determined based on the discharges in the main channel and left and right overbank segments of the reach (see Equation (3)).

Energy Loss Coefficients

Several types of loss coefficients are utilized by the program to evaluate energy losses: (1) Manning's n values or equivalent roughness " k " values for friction loss, (2) contraction and expansion coefficients to evaluate transition (shock) losses, and (3) bridge and culvert loss coefficients to evaluate losses related to weir shape, pier configuration, pressure flow, and entrance and exit conditions. Energy loss coefficients associated with bridges and culverts are discussed in "Modeling Bridges" and "Modeling Culverts" of the HEC-RAS Hydraulic Reference manual (<https://www.hec.usace.army.mil/confluence/rasdocs>).

Manning's n . Selection of an appropriate value for Manning's n is very significant to the accuracy of the computed water surface elevations. The value of Manning's n is highly variable and depends on a number of factors including: surface roughness; vegetation;

channel irregularities; channel alignment; scour and deposition; obstructions; size and shape of the channel; stage and discharge; seasonal changes; temperature; and suspended material and bedload.

In general, Manning's n values should be calibrated whenever observed water surface elevation information (gaged data, as well as high water marks) is available. When gaged data are not available, values of n computed for similar stream conditions or values obtained from experimental data should be used as guides in selecting n values.

There are several references a user can access that show Manning's n values for typical channels. An extensive compilation of n values for streams and floodplains can be found in Chow's book "Open- Channel Hydraulics" [Chow, 1959]. Excerpts from Chow's book, for the most common types of channels, are shown in Table 3-1 below. Chow's book presents additional types of channels, as well as pictures of streams for which n values have been calibrated.

Table 3-1 Manning's n Values

Type of Channel and Description	Minimum	Normal
<i>A. Natural Streams</i>		
1. Main Channels		
a. Clean, straight, full, no rifts or deep pools	0.025	0.030
b. Same as above, but more stones and weeds	0.030	0.035
c. Clean, winding, some pools and shoals	0.033	0.040
d. Same as above, but some weeds and stones	0.035	0.045
e. Same as above, lower stages, more ineffective slopes and sections	0.040	0.048
f. Same as "d" but more stones	0.045	0.050
g. Sluggish reaches, weedy. deep pools	0.050	0.070
h. Very weedy reaches, deep pools, or floodways with heavy stands of timber and brush	0.070	0.100
2. Flood Plains		
a. Pasture no brush		
1. Short grass	0.025	0.030
2. High grass	0.030	0.035
b. Cultivated areas		

1. No crop	0.020	0.030
2. Mature row crops	0.025	0.035
3. Mature field crops	0.030	0.040
c. Brush		
1. Scattered brush, heavy weeds	0.035	0.050
2. Light brush and trees, in winter	0.035	0.050
3. Light brush and trees, in summer	0.040	0.060
4. Medium to dense brush, in winter	0.045	0.070
5. Medium to dense brush, in summer	0.070	0.100
d. Trees		
1. Cleared land with tree stumps, no sprouts	0.030	0.040
2. Same as above, but heavy sprouts	0.050	0.060
3. Heavy stand of timber, few down trees, little undergrowth, flow below branches	0.080	0.100
4. Same as above, but with flow into branches	0.100	0.120
5. Dense willows, summer, straight	0.110	0.150

3. Mountain Streams, no vegetation in channel, banks usually steep, with trees and brush on banks submerged

a. Bottom: gravels, cobbles, and few boulders	0.030	0.040
b. Bottom: cobbles with large boulders	0.040	0.050

B. Lined or Built-Up Channels

1. Concrete

a. Trowel finish	0.011	0.013
b. Float Finish	0.013	0.015
c. Finished, with gravel bottom	0.015	0.017
d. Unfinished	0.014	0.017
e. Gunite, good section	0.016	0.019

f. Gunite, wavy section	0.018	0.022
g. On good excavated rock	0.017	0.020
h. On irregular excavated rock	0.022	0.027
2. Concrete bottom float finished with sides of:		
a. Dressed stone in mortar	0.015	0.017
b. Random stone in mortar	0.017	0.020
c. Cement rubble masonry, plastered	0.016	0.020
d. Cement rubble masonry	0.020	0.025
e. Dry rubble on riprap	0.020	0.030
3. Gravel bottom with sides of:		
a. Formed concrete	0.017	0.020
b. Random stone in mortar	0.020	0.023
c. Dry rubble or riprap	0.023	0.033
4. Brick		
a. Glazed	0.011	0.013
b. In cement mortar	0.012	0.015

5. Metal

a. Smooth steel surfaces	0.011	0.012
b. Corrugated metal	0.021	0.025

6. Asphalt

a. Smooth	0.013	0.013
b. Rough	0.016	0.016

7. Vegetal lining 0.030

C. Excavated or Dredged Channels

1. Earth, straight and uniform

a. Clean, recently completed	0.016	0.018
b. Clean, after weathering	0.018	0.022
c. Gravel, uniform section, clean	0.022	0.025
d. With short grass, few weeds	0.022	0.027

2. Earth, winding and sluggish

a. No vegetation	0.023	0.025
b. Grass, some weeds	0.025	0.030
c. Dense weeds or aquatic plants in deep channels	0.030	0.035
d. Earth bottom and rubble side	0.028	0.030
e. Stony bottom and weedy banks	0.025	0.035
f. Cobble bottom and clean sides	0.030	0.040

3. Dragline-excavated or dredged

a. No vegetation	0.025	0.028
b. Light brush on banks	0.035	0.050

4. Rock cuts

a. Smooth and uniform	0.025	0.035
b. Jagged and irregular	0.035	0.040

5. Channels not maintained, weeds and brush

a. Clean bottom, brush on sides	0.040	0.050
b. Same as above, highest stage of flow	0.045	0.070
c. Dense weeds, high as flow depth	0.050	0.080
d. Dense brush, high stage	0.080	0.100

Other sources that include pictures of selected streams as a guide to n value determination are available (Fasken, 1963; Barnes, 1967; and Hicks and Mason, 1991). In general, these references provide colour photos with tables of calibrated n values for a range of flows.

Although there are many factors that affect the selection of the n value for the channel, some of the most important factors are the type and size of materials that compose the bed and banks of a channel, and the shape of the channel. Cowan (1956) developed a procedure for estimating the effects of these factors to determine the value of Manning's n of a channel. In **Cowan's procedure**, the value of n is computed by the following equation:

$$n = (n_0 + n_1 + n_2 + n_3 + n_4)m$$

Symbol	Description
n_b	Base value for n for a straight uniform, smooth channel in natural materials
n_1	Value added to correct for surface irregularities
n_2	Value for variations in shape and size of the channel
n_3	Value for obstructions
n_4	Value for vegetation and flow conditions
m	Correction factor to account for meandering of the channel

A detailed description of Cowan's method can be found in "Guide for Selecting Manning's Roughness Coefficients for Natural Channels and Flood Plains" (FHWA, 1984). This report was developed by the U.S. Geological Survey (Arcement, 1989) for the Federal Highway Administration. The report also presents a method similar to Cowan's for developing Manning's n values for flood plains, as well as some additional methods for densely vegetated flood plains.

Limerinos (1970) related n values to hydraulic radius and bed particle size based on samples from 11 stream channels having bed materials ranging from small gravel to medium size boulders. The Limerinos equation is as follows:

$$n = \frac{(0.0926)R^{1/6}}{1.16 + 201 \log \left(\frac{R}{d_{84}} \right)}$$

Symbol	Description
R	Hydraulic radius, in feet (data range was 1.0 to 6.0 feet)
d_{84}	Particle diameter, in feet, that equals or exceeds that of 84 percent of the particles (data range was 1.5 mm to 250 mm)

The Limerinos equation fit the data that he used very well, in that the coefficient of correlation $R^2 = 0.88$ and the standard error of estimates for values of $n/R^{1/6} = 0.0087$. Limerinos selected reaches that had a minimum amount of roughness, other than that caused by the bed material. The Limerinos equation provides a good estimate of the base n value. The base n value should then be increased to account for other factors, as shown above in Cowen's method.

Jarrett (1984) developed an equation for high gradient streams (slopes greater than 0.002). Jarrett performed a regression analysis on 75 data sets that were surveyed from 21 different streams. Jarrett's equation for Manning's n is as follows:

$$n = 0.39S^{0.38} R^{-0.16}$$

Symbol	Description
S	The friction slope. The slope of the water surface can be used when the friction slope is unknown.
R	The Hydraulic Radius of the main channel at bank full flow.

Jarrett (1984) states the following limitations for the use of his equation:

1. The equations are applicable to natural main channels having stable bed and bank materials (gravels, cobbles, and boulders) without backwater.
2. The equations can be used for slopes from 0.002 to 0.04 and for hydraulic radii from 0.5 to 7.0 feet (0.15 to 2.1 m). The upper limit on slope is due to a lack of verification data available for the slopes of high-gradient streams. Results of the regression analysis indicate that for hydraulic radius greater than 7.0 feet (2.1 m), n did not vary significantly with depth; thus extrapolating to larger flows should not be too much in error as long as the bed and bank material remain fairly stable.
3. During the analysis of the data, the energy loss coefficients for contraction and expansion were set to 0.0 and 0.5, respectively.
4. Hydraulic radius does not include the wetted perimeter of bed particles.
5. These equations are applicable to streams having relatively small amounts of suspended sediment.

Because Manning's n depends on many factors such as the type and amount of vegetation, channel configuration, stage, etc., several options are available in HEC-RAS to vary n . When three n values are sufficient to describe the channel and overbanks, the user can enter the three n values directly onto the cross-section editor for each cross section. Any of the n values

may be changed at any cross section. Often three values are not enough to adequately describe the lateral roughness variation in the cross section; in this case the “Horizontal Variation of n Value” should be selected from the “Options” menu of the cross-section editor. If n values change within the channel, the criterion described in "Theoretical Basis for One-Dimensional and Two-Dimensional Hydrodynamic Calculations", under composite n values, is used to determine whether the n values should be converted to a composite value using Equation (5).

Equivalent Roughness “k”. An equivalent roughness parameter “k”, commonly used in the hydraulic design of channels, is provided as an option for describing boundary roughness in HEC-RAS. Equivalent roughness, sometimes called “roughness height,” is a measure of the linear dimension of roughness elements, but is not necessarily equal to the actual, or even the average, height of these elements. In fact, two roughness elements with different linear dimensions may have the same “k” value because of differences in shape and orientation [Chow, 1959].

The advantage of using equivalent roughness “k” instead of Manning’s “n” is that “k” reflects changes in the friction factor due to stage, whereas Manning’s “n” alone does not. This influence can be seen in the definition of Chezy’s “C” (English units) for a rough channel (Equation 2-6, USACE, 1991):

$$C = 32.6 \log_{10} \left[\frac{12.2 R}{k} \right]$$

Symbol	Description
<i>C</i>	Chezy roughness coefficient
Symbol	Description
<i>R</i>	hydraulic radius
<i>k</i>	equivalent roughness

Note that as the hydraulic radius increases (which is equivalent to an increase in stage), the friction factor “C” increases. In HEC-RAS, “k” is converted to a Manning’s “n” by using the above equation and equating the Chezy and Manning’s equations (Equation 2-4, USACE, 1991) to obtain the following:

English Units:

$$n = \frac{1.486 R^{1/6}}{32.6 \log_{10} \left[12.2 \frac{R}{k} \right]}$$

$$n = \frac{R^{1/6}}{18 \log_{10} \left[12.2 \frac{R}{k} \right]}$$

Symbol	Description
<i>n</i>	Manning’s roughness coefficient

Again, this equation is based on the assumption that all channels (even concrete-lined channels) are “hydraulically rough.” A graphical illustration of this conversion is available

[USACE, 1991]. Horizontal variation of “k” values is described in the same manner as horizontal variation of Manning's “n” values. See "Modeling Culverts" of the HEC-RAS User's Manual and Hydraulic Reference Manual, to learn how to enter k values into the program. Up to twenty values of “k” can be specified for each cross section. Tables and charts for determining “k” values for concrete-lined channels are provided in EM 1110-2-1601 [USACE, 1991]. Values for riprap-lined channels may be taken as the theoretical spherical diameter of the median stone size. Approximate “k” values [Chow, 1959] for a variety of bed materials, including those for natural rivers are shown in Table 3-2.

Table 3-2 Equivalent Roughness Values of Various Bed Materials

	k (Feet)
Brass, Cooper, Lead, Glass	0.0001 - 0.0030
Wrought Iron, Steel	0.0002 - 0.0080
Asphalted Cast Iron	0.0004 - 0.0070
Galvanized Iron	0.0005 - 0.0150
Cast Iron	0.0008 - 0.0180
Wood Stave	0.0006 - 0.0030
Cement	0.0013 - 0.0040
Concrete	0.0015 - 0.0100
Drain Tile	0.0020 - 0.0100
Riveted Steel	0.0030 - 0.0300
Natural River Bed	0.1000 - 3.0000

The values of "k" (0.1 to 3.0 ft.) for natural river channels are normally much larger than the actual diameters of the bed materials to account for boundary irregularities and bed forms.

Contraction and Expansion Coefficients. Contraction or expansion of flow due to changes in the cross section is a common cause of energy losses within a reach (between two cross sections). Whenever this occurs, the loss is computed from the contraction and expansion coefficients specified on the cross-section data editor. The coefficients, which are applied between cross sections, are specified as part of the data for the upstream cross section. The coefficients are multiplied by the absolute difference in velocity heads between the current cross section and the next cross section downstream, which gives the energy loss caused by the transition (Equation (2) of "Theoretical Basis for One-Dimensional and Two-Dimensional Hydrodynamic Calculations"). Where the change in river cross section is small, and the flow is subcritical, coefficients of contraction and expansion are typically on the order of 0.1 and 0.3, respectively. When the change in effective cross section area is abrupt such as at bridges, contraction and expansion coefficients of 0.3 and 0.5 are often used. On occasion, the coefficients of contraction and expansion around bridges and culverts may be as high as 0.6 and 0.8, respectively. These values may be changed at any cross section. For additional information concerning transition losses and for information on bridge loss coefficients, see "Modeling Bridges" in the Hydraulic Reference Manual. Typical values for contraction and expansion coefficients, for subcritical flow, are shown in Table 3-3 below.

Table 3-3 Subcritical Flow Contraction and Expansion Coefficients

	Contraction	Expansion
No transition loss computed	0.0	0.0
Gradual transitions	0.1	0.3
Typical Bridge sections	0.3	0.5
Abrupt transitions	0.6	0.8

The maximum value for the contraction and expansion coefficient is one (1.0).

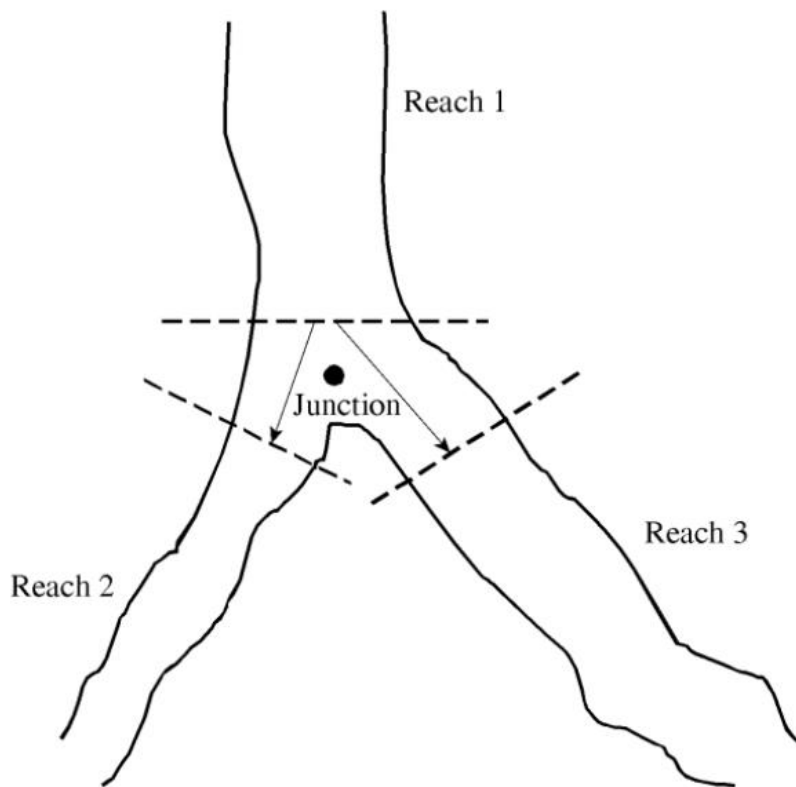
Note

In general, the empirical contraction and expansion coefficients should be lower for supercritical flow.

In supercritical flow the velocity heads are much greater, and small changes in depth can cause large changes in velocity head. Using contraction and expansion coefficients that would be typical for subcritical flow can result in over estimation of the energy losses and oscillations in the computed water surface profile. In constructed trapezoidal and rectangular channels, designed for supercritical flow, the user should set the contraction and expansion coefficients to zero in the reaches where the cross-sectional geometry is not changing shape. In reaches where the flow is contracting and expanding, the user should select contraction and expansion coefficients carefully. Typical values for gradual transitions in supercritical flow would be around 0.01 for the contraction coefficient and 0.03 for the expansion coefficient. As the natural transitions begin to become more abrupt, it may be necessary to use higher values, such as 0.05 for the contraction coefficient and 0.2 for the expansion coefficient. If there is no contraction or expansion, the user may want to set the coefficients to zero for supercritical flow.

Stream Junction Data

Stream junctions are defined as locations where two or more streams come together or split apart. Junction data consists of reach lengths across the junction and tributary angles (only if the momentum equation is selected). Reach lengths across the junction are entered in the Junction Data editor. This allows for the lengths across very complicated confluences (e.g., flow splits) to be accommodated. An example of this is shown in the figure below.



As shown in the figure above, using downstream reach lengths, for the last cross section in Reach 1, would not adequately describe the lengths across the junction. It is therefore necessary to describe lengths across junctions in the Junction Data editor. For the example shown in the figure above, two lengths would be entered. These lengths should represent the average distance that the water will travel from the last cross section in Reach 1 to the first cross section of the respective reaches. In general, the cross sections that bound a junction should be placed as close together as possible. This will minimize the error in the calculation of energy losses across the junction. In HEC-RAS, for steady flow hydraulic computations, a junction can be modelled by either the energy equation (Equation (1) of "Theoretical Basis for One-Dimensional and Two-Dimensional Hydrodynamic Calculations") or the momentum equation. The energy equation does not take into account the angle of any tributary coming in or leaving the main stream, while the momentum equation does. In most cases, the amount of energy loss due to the angle of the tributary flow is not significant, and using the energy equation to model the junction is more than adequate. However, there are situations where the angle of the tributary can cause significant energy losses. In these situations, it would be more appropriate to use the momentum approach. When the momentum approach is selected, an angle for all tributaries of the main stem must be entered. A detailed description of how junction calculations are made can be found in "Overview of Optional Capabilities" of the Hydraulic Reference Manual. For unsteady flow computations, HEC-RAS has two options for the hydraulic computations at a junction. The default option is a very simple assumption that the water surface computed at the downstream side of a flow combining junction, is used for the cross sections just upstream of the junction. If this is not a good assumption (such as for steeper river systems), there is an option to perform an energy balance across the junction in order to compute the upstream water surface elevations.

Steady Flow Data

Steady flow data are required in order to perform a steady water surface profile calculation. Steady flow data consist of: flow regime; boundary conditions; and discharge information (peak flows or flow data from a specific instance in time).

Flow Regime

Profile computations begin at a cross section with known or assumed starting conditions and proceed upstream for subcritical flow or downstream for supercritical flow. The flow regime (subcritical, supercritical, or mixed flow regime) is specified on the Steady Flow Analysis window of the user interface. Subcritical profiles computed by the program are constrained to critical depth or above, and supercritical profiles are constrained to critical depth or below. In cases where the flow regime will pass from subcritical to supercritical, or supercritical to subcritical, the program should be run in a mixed flow regime mode. For a detailed discussion of mixed flow regime calculations, see "Overview of Optional Capabilities" of the Hydraulic Reference Manual.

Boundary Conditions

Boundary conditions are necessary to establish the starting water surface at the ends of the river system (upstream and downstream). A starting water surface is necessary in order for the program to begin the calculations. In a subcritical flow regime, boundary conditions are only necessary at the downstream ends of the river system. If a supercritical flow regime is going to be calculated, boundary conditions are only necessary at the upstream ends of the river system. If a mixed flow regime calculation is going to be made, then boundary conditions must be entered at all ends of the river system. The boundary conditions editor contains a table listing every reach. Each reach has an upstream and a downstream boundary condition. Connections to junctions are considered internal boundary conditions. Internal boundary conditions are automatically listed in the table, based on how the river system was defined in the geometric data editor. The user is only required to enter the necessary external boundary conditions. There are four types of boundary conditions available to the user:

Known Water Surface Elevations - For this boundary condition the user must enter a known water surface elevation for each of the profiles to be computed.

Critical Depth - When this type of boundary condition is selected, the user is not required to enter any further information. The program will calculate critical depth for each of the profiles and use that as the boundary condition.

Normal Depth - For this type of boundary condition, the user is required to enter an energy slope that will be used in calculating normal depth (using Manning's equation) at that location. A normal depth will be calculated for each profile based on the user-entered slope. In general, the energy slope can be approximated by using the average slope of the channel, or the average slope of the water surface in the vicinity of the cross section.

Rating Curve - When this type of boundary condition is selected, a pop-up window appears allowing the user to enter an elevation versus flow rating curve. For each profile, the elevation is interpolated from the rating curve given the flow, using linear interpolation between the user-entered points. Whenever the water surface elevations at the boundaries of the study are unknown; and a user defined water surface is required at the boundary to start the calculations; the user must either estimate the water surface, or select normal depth or critical depth. Using an estimated water surface will incorporate an error in the water surface

profile in the vicinity of the boundary condition. If it is important to have accurate answers at cross sections near the boundary condition, additional cross sections should be added. If a subcritical profile is being computed, then additional cross sections need only be added below the downstream boundaries. If a supercritical profile is being computed, then additional cross sections should be added upstream of the relevant upstream boundaries. If a mixed flow regime profile is being computed, then cross sections should be added upstream and downstream of all the relevant boundaries. In order to test whether the added cross sections are sufficient for a particular boundary condition, the user should try several different starting elevations at the boundary condition, for the same discharge. If the water surface profile converges to the same answer, by the time the computations get to the cross sections that are in the study area, then enough sections have been added, and the boundary condition is not affecting the answers in the study area.

Discharge Information

Discharge information is required at each cross section in order to compute the water surface profile. Discharge data are entered from upstream to downstream for each reach. At least one flow value must be entered for each reach in the river system. Once a flow value is entered at the upstream end of a reach, it is assumed that the flow remains constant until another flow value is encountered with the same reach. The flow rate can be changed at any cross section within a reach. However, the flow rate cannot be changed in the middle of a bridge, culvert, or stream junction. Flow data must be entered for the total number of profiles that are to be computed.

Unsteady Flow Data

Unsteady flow data are required in order to perform an unsteady flow analysis. Unsteady flow data consists of boundary conditions (external and internal), as well as initial conditions.

Boundary Conditions

Boundary conditions must be established at all of the open ends of the river system being modelled. Upstream ends of a river system can be modelled with the following types of boundary conditions: flow hydrograph (most common upstream boundary condition); stage hydrograph; flow and stage hydrograph. Downstream ends of the river system can be modelled with the following types of boundary conditions: rating curve, normal depth (Manning's equation); stage hydrograph; flow hydrograph; stage and flow hydrograph. Boundary conditions can also be established at internal locations within the river system. The user can specify the following types of boundary conditions at internal cross sections: lateral inflow hydrograph; uniform lateral inflow hydrograph; groundwater interflow; and Internal Stage and flow hydrograph. Additionally, any gated structures that are defined within the system (inline, lateral, or between storage areas and/or 2D flow areas) could have the following types of boundary conditions in order to control the gates: time series of gate openings; elevation-controlled gate; navigation dam; Rules; or internal observed stage and flow.

Initial Conditions

In addition to boundary conditions, the user is required to establish the initial conditions (flow and stage) at all nodes in the system at the beginning of the simulation. Initial conditions can be established in two different ways. The most common way is for the user to enter flow data

for each reach, and then have the program compute water surface elevations by performing a steady flow backwater analysis. A second method can only be done if a previous run was made. This method allows the user to write a file of flow and stage from a previous run, which can then be used as the initial conditions for a subsequent run.

In addition to establishing the initial conditions within the river system, the user must define the starting water surface elevation in any storage areas and 2D flow area that are defined. This is accomplished from the initial conditions editor. The user can enter a stage for each storage area within the system. 2D Flow areas have several ways of establishing initial conditions within the 2D flow area.

For more information on unsteady flow data, please review "Modeling Multiple Bridge and/or Culvert Openings" of the HEC-RAS User's manual and the Hydraulic Reference Manual.